

Sponsored Search and Product Market Concentration*

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This paper examines how sponsored search auctions shape product market competition. I develop a model in which firms bid for prominence on a platform and set prices, under a cost-per-click commission model. Consumers observe prices before search but incur costs to learn match values. The analysis reveals a novel mechanism: ad commissions pass through asymmetrically to prices. For the prominent firm, commissions function like a fixed cost, leaving its price largely unaffected; for the non-prominent firm, they act as a marginal cost, raising its price. This asymmetry reduces competition and can even reverse price rankings. I explore alternative commission models and provide suggestive empirical evidence consistent with the theory.

Keywords: Digital Economy, Market Power, Ad Auction, Consumer Search.

JEL Codes: L1, D4, D82, D83

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1 Introduction

The rise of dominant digital platforms has been accompanied by concerns of declining product market competition across sectors.¹ This contrasts with early expectations that the Internet would foster highly competitive markets by reducing search costs and enabling more firms to reach consumers.² In such online markets, firms often compete to advertise their products and secure prominent positions to reach consumers. How do these sponsored advertisements affect firms' pricing and, consequently, price competition? Given that platforms' business models frequently rely on commissions from selling these advertisements,³ do platforms' commission structures alleviate or exploit consumer search frictions, and what are the implications for overall welfare? Understanding the effect of sponsored ads on product markets is therefore crucial for promoting competitive online markets.

While competition *among* platforms has received extensive policy attention, understanding competition *within* a platform is equally important for designing effective regulation.⁴ Existing studies on within-platform competition have largely focused on strategic interactions between firms and the platform.⁵ However, consumer behaviour remains an understudied determinant of digital market outcomes. In this paper, I examine a key mechanism through which platforms influence these outcomes – position auctions – and its interaction with consumer search frictions to understand firms' pricing strategies.

I model the platform's ad revenue as a function of an auction that it designs and conducts to determine (i) the order in which ads (of firms) are displayed, and (ii) the ad commission to be paid at each position. I show that this ordered display not only determines consumers' click-through and firms' pricing behaviour, but also endogenously affects product market concentration and platform revenue. Specifically, I highlight a novel channel - the asymmetry, across ad positions, in pass-through from ad commissions to product prices - which leads to an increase in product market concentration and a decrease in consumer surplus. I also provide novel empirical evidence, using data from an anonymous digital platform in the US, that provides support to this channel. Importantly, this channel persists even when product prices are fully observable, and even when the intermediary does not compete directly with firms, indicating that auction design combined with search frictions in learning product attributes alone can exacerbate market

¹See, for instance, [Khan \(2018\)](#), [ACCC \(2019\)](#), and [CMA \(2020b\)](#) for evidence on platform dominance, and [De Loecker, Eeckhout, and Unger \(2020\)](#) for trends in concentration.

²See, for instance, [Brynjolfsson and Smith \(2000\)](#) and [Brynjolfsson, Hu, and Smith \(2003\)](#) for early empirical evidence, and [Pozzi \(2012\)](#) for later counter-evidence.

³For example, Google ad revenue accounts for more than 80% of Alphabet's total revenue, and Amazon reports annual ad revenue growth exceeding 40% (see [Google Investor Relations](#) and [Amazon Investor Relations](#)).

⁴See, for instance, [Broadbent \(2020\)](#) for an overview of the Digital Markets Act and the Digital Services Act.

⁵See, for instance, [Caffarra, Etro, Latham, and Scott-Morton \(2020\)](#) for policy discussion and [Belleflamme and Peitz \(2019\)](#), [Teh \(2022\)](#) for recent academic work.

concentration.

I build a model of the e-commerce environment comprising of three key features. First, product prices are more easily accessible to consumers than product characteristics.⁶ For example, when consumers see a list of ads, they observe the prices of the products on the ads immediately. This information is obtained at a relatively much lower cost compared to *searching* (clicking on each ad) to learn the *match value* from product characteristics/attributes.

Second, prominence strongly influences consumer behavior. Consumers often start with the top-listed ad or one occupying a larger display space.⁷ In equilibrium, I endogenize prominence and show that this behaviour arises naturally, consistent with the rationale that consumers begin with the lower-priced firm.

Third, the platform allocates positions through a Generalized Second Price (GSP) auction to maximize revenue.⁸ The platform's revenue equals cost per click (CPC) times the number of clicks, and the auction determines both the display order and CPC payments, which depend on position and market competition.⁹

Using this framework, I determine the endogenous search and demand behaviour of consumers over products (or ads, in this paper), as well as the pricing decisions of firms (for their product) and the platform (for its ad position). A product's price is determined by both consumer preferences and the pass-through from the CPC paid to the platform, which in turn depends on its display position. Consequently, click volumes and CPC vary across positions. These differences allow me to micro-found the 'value of a click' based on position and firm characteristics, thus avoiding the common assumption of position homogeneity.¹⁰

The analysis proceeds in two steps. First, I examine market outcomes without an auction, assuming consumers start from the prominent firm, in order to disentangle the effect of prominence and price visibility on product pricing and competition from the effect of position auction. I show that, when prices are observable or posted, firm 1 charges a higher price than firm 2 ($p_1 > p_2$), and this dispersion grows with search cost. Hence, with posted prices, a lower p_2 not only retains more visitors, but is also likely to attract more of them. The latter effect makes

⁶See Figure A.1 for examples and Gorodnichenko and Talavera (2017) for discussion on price transparency online.

⁷See, for instance, Granka, Joachims, and Gay (2004) for eye-tracking evidence, Ghose and Yang (2009) for evidence from panel data at Google, Agarwal, Hosanagar, and Smith (2011) for evidence from a field experiment at Google, Narayanan and Kalyanam (2015) for evidence from an RDD approach, Ursu (2018) for evidence from a field experiment at Expedia, Simonov and Hill (2021) for quasi-experimental evidence from Bing, and Moshary (2021) for evidence from a field experiment.

⁸This is one of the most common auction formats employed by digital platforms to determine ad positions. See, for instance, Edelman, Ostrovsky, and Schwarz (2007) and Varian (2007).

⁹This formulation of platform revenue is motivated by Google ranking firms based on CPC times the estimated click-through rate. See <https://support.google.com/google-ads/answer/142918?hl=en> (Accessed May 14, 2020).

¹⁰See, for instance, Goldman and Rao (2016) for evidence from Bing on heterogeneous effect of position on click-through rates.

firm 2's demand more elastic and puts it at a competitive disadvantage.

Second, I introduce endogenous CPC payments through a GSP auction. The platform sets the reserve price to maximize revenue, subject to firms' participation and incentive constraints. In this full equilibrium, consumers choose their search order, and for some search costs, prices increase with order ($p_1 < p_2$), consistent with directed search, where consumers indeed start their search from the lower-priced prominent firm. However, the order in my model is driven by the CPC structure: for the prominent firm, CPC acts like a fixed cost (clicks are insensitive to price), while for the non-prominent firm, CPC behaves like a marginal cost (clicks depend on price). This asymmetry forces firm 2 to raise p_2 , further weakening its competitive position.

The intuition for this result is grounded in the *conversion rate* (buyers-to-visitors ratio), which captures price elasticity. This ratio is determined by the price elasticity of consumers that a firm faces. In standard prominence models, the prominent firm faces more elastic demand ("fresh" consumers, as referred to in the literature) and charges a lower price, creating a self-fulfilling order. In my setting, price visibility breaks this mechanism, but the auction restores it: firms compete for positions to secure conversion rates that minimize CPC pass-through. This cost-side channel closes the loop and generates equilibrium prices that "self-fulfill" prominence. Comparative statics with search cost yield falsifiable predictions on conversion rates and pricing, which I test empirically in Section 5.

In the empirical analysis, I use clickstream data from a large U.S. platform to assess whether the model's key predictions are reflected in the data. The correlational exercise focuses on two outcomes central to the theoretical mechanism: (i) variation in CPC across positions and its implications for pricing incentives, and (ii) convergence in conversion rates as an indicator of symmetric pass-through. The results show that ads in prominent positions pay higher CPC and receive more clicks, while conversion rates remain similar across positions. These patterns align with the model's prediction that pricing asymmetry is driven by CPC pass-through rather than differences in product quality or consumer preferences.

Finally, as an extension, I incorporate vertical differentiation to examine whether platform recommendations align with consumer welfare. While misalignment is possible, since the platform's revenue depends on search volume, results show that higher-quality firms endogenously secure prominent positions because they generate higher CPC revenue for the platform.¹¹ I also compare welfare under alternative intermediary revenue models. Replacing CPC with Cost Per Purchase (CPP) or introducing consumer subscription fees improves consumer surplus but reduces intermediary revenue. A combined CPC plus subscription model induces cross-subsidization, reflecting a trade-off between ad revenue and consumer fees. I also show that a

¹¹This is in contrast with the case where prices are unposted; see Jerath, Ma, Park, and Srinivasan (2011).

Generalized First Price auction yields outcomes similar to GSP in the duopoly model.

1.1 Literature Review

This paper contributes to three broad strands of literature, namely those of firm-intermediary interaction, ordered consumer search, and advertising.

Firm-intermediary interaction. A foundational literature studies position auctions as a mechanism for allocating sponsored ads. [Edelman, Ostrovsky, and Schwarz \(2007\)](#) and [Varian \(2007\)](#) analyze Generalized Second Price (GSP) auctions and characterize bidding equilibria. Subsequent work explores auction design and efficiency under incomplete information ([Athey and Ellison, 2011](#), [Chen and He, 2011](#)). Recent contributions examine strategic implications of auction formats and platform design: [Anderson and Renault \(2021\)](#) highlights demand-side externalities, while [Kang \(2021\)](#) and [Motta and Penta \(2022\)](#) study alternative pricing rules and their competitive effects. Closer to my work, [Armstrong and Zhou \(2011\)](#) examines how cost per purchase or lump sum commission payments inflate suppliers' marginal costs and distort retail prices.

These papers abstract from consumer search frictions and price visibility, features that are prevalent in digital platforms and central to my analysis. By embedding ordered search into a position-auction framework, I microfound the heterogeneous value of clicks and show how cost-per-click (CPC) payments distort pricing incentives by generating asymmetric cost pass-through across positions, a mechanism absent in prior models.

Ordered consumer search. Classic models of prominence, such as [Armstrong, Vickers, and Zhou \(2009\)](#), show that firms listed first attract more visitors and typically charge lower prices when prices are costly to observe. Extensions consider heterogeneous search costs ([Arbatskaya, 2007](#)) and directed search ([Choi, Dai, and Kim, 2018](#), [Haan, Moraga-González, and Petrikaitė, 2018](#)). Closer to my work, [The and Wright \(2022\)](#) examines searching consumers, sponsored positions, and pricing in a setting with specific matching rules, and [Obradovits and Plaickner \(2023\)](#) analyzes price-directed search, where ranking depends on posted prices.

My paper differs in three fundamental ways. First, search order is determined endogenously through an auction rather than exogenously by prices, introducing a two-stage game and strategic bidding. Second, cost-per-click (CPC) payments create a novel channel of asymmetry: CPC acts as a fixed cost for the prominent firm and a marginal cost for the non-prominent firm. Third, this mechanism reverses standard predictions – that is prominence combined with CPC can lead to higher prices and greater concentration – in contrast to the price compression typically found in price-directed search models. I also provide welfare analysis under alternative platform revenue models, highlighting trade-offs between consumer surplus and market concentration.

Advertising. A large literature debates whether advertising expands consideration sets or

distorts consumer choice. [Sahni and Zhang \(2019\)](#) and [Moshary \(2021\)](#) emphasize the informative role of ads, while recent work highlights persuasive effects and intermediary bias ([Heidhues and Kőszegi, 2018](#), [Jullien and Sand-Zantman, 2021](#)). My analysis shows that persuasive outcomes can arise endogenously: auction design and CPC pass-through create pricing distortions that restrict consumer choice, even when platforms rank firms by relevance. Extensions to asymmetric firms further reveal that advertising can act as a coordination device, reinforcing the advantage of high-value firms.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 analyzes pricing under prominence without auctions. Section 4 introduces the auction mechanism and derives equilibrium outcomes. Section 5 provides empirical evidence. Section 6 concludes with policy implications and directions for future research. Proofs are provided in the appendix.

2 Model

Consider a market with one intermediary (m), two firms (i, j), and a unit mass of consumers. Each firm produces a single product and seeks to display one sponsored ad through the intermediary.¹² Firms and the intermediary maximize their own profits, while consumers maximize utility. Numeric subscripts denote position: 1 indicates the prominent position and 2 the non-prominent position.

The intermediary runs a Generalized Second Price (GSP) auction to determine the order in which firms are listed on its platform. It sets the auction reserve price $\hat{b} \in \mathbb{R}_+$ to maximize its revenue. Revenue equals the cost per click (CPC) paid by each firm multiplied by the number of clicks that firm receives. The platform does not observe consumers' individual valuations ex ante.

Firms compete for prominence by bidding the CPC they are willing to pay. Positions are assigned based on the auction outcome. After positions are revealed, each firm sets its product price (see Appendix Section B for discussion on timing). Let $b_i, b_j \in \mathbb{R}_+$ denote bids and $p_i, p_j \in \mathbb{R}_+$ denote prices. Let $r_i(b_i, b_j, \hat{b}), r_j(b_i, b_j, \hat{b}) \in \mathbb{R}_+$ denote the CPC paid by each firm to the intermediary. Bids are inputs to the auction; CPC payments are outputs. For simplicity, marginal production costs are normalized to zero.

Consumers face a search cost $s \in \mathbb{R}_+$.¹³ Consistent with common practice on e-commerce

¹²Throughout, I use the terms 'firm' and 'product' interchangeably to refer to its ad.

¹³As consumers pay a uniform search cost in my model, any differences in costly-to-observe "add-on" prices are inconsequential in equilibrium, following the Chicago-style arguments ([Lal and Matutes, 1994](#)). Moreover, add-ons (e.g., shipping cost) are often used to recover market-specific (e.g., product/location-based) fixed costs ([Ellison, 2005](#)) and are independent of total visitors. Therefore, one could also consider "add-ons" as a part of the match value, in my model, that is discovered after search. Hence, my framework allows me to focus on the firms' choice

platforms, I assume that consumers observe the order of listed firms and the prices of each product without incurring the search cost.¹⁴ However, consumers learn their idiosyncratic valuation of a product only after visiting the seller’s page, thereby paying the search cost. For example, consider a website where prices are displayed on the homepage, but product-specific features are revealed only upon clicking the ad (see Figure A.1 for examples).

Let v_i, v_j denote a consumer’s idiosyncratic valuations for products offered by firms i and j , respectively. The outside option for consumers is normalized to zero, and recall is assumed to be costless. Consumers draw valuations from an identical distribution F :

$$v_i, v_j \sim F[\underline{v}, \bar{v}],$$

where F is twice differentiable and $f(\cdot) > 0$ on its support. Consumer utility from purchasing product i is given by $u(p_i) = v_i - p_i$. Firms and the intermediary are assumed to be risk-neutral, with an outside option of zero.

The order of events is as follows: First, the intermediary sets the auction reserve price. Second, firms place their bids. Third, positions are revealed. Fourth, firms set product prices. Finally, consumers make search and purchase decisions. The solution concept is Subgame Perfect Nash Equilibrium.

The sequential timing reflects the institutional features of online advertising. Platform auction formats change very rarely, while CPC bids are adjusted by firms at a higher frequency (Edelman, Ostrovsky, and Schwarz, 2007; Varian, 2007). The key assumption that product prices are set conditional on position is supported by the widespread use of algorithmic repricing software tools that enable sellers to re-optimize prices in real time as a function of their display ranking (Chen, Mislove, and Wilson, 2016).¹⁵ Robustness to alternative timing assumptions are discussed in Online Appendix Section E.

3 Consumer and Firm Behaviour

I begin by analyzing the effect of consumer search frictions on firms’ pricing strategies when positions are exogenous and there are no cost-per-click (CPC) payments. Section 4 then introduces the intermediary, which endogenizes positions and CPC costs.

of the advertised product price.

¹⁴I compare these results with a benchmark model where prices are not visible prior to search in Online Appendix, Section A.

¹⁵Chen, Mislove, and Wilson (2016) document that sellers on an e-commerce platforms, namely Amazon in their study, may change prices tens or hundreds of times per day, with pricing strategies explicitly designed to exploit their position allocation.

3.1 Consumer Search

To capture the notion of prominence in a simple yet stark manner, I assume that the first search (click) is free for consumers, while a search cost $s > 0$ applies for visiting firm 2. This assumption aligns with empirical evidence from online environments, where consumers typically begin with the most salient or top-ranked option.¹⁶ Moreover, as shown in Appendix Section B, this assumption is not the key driver of the intuition behind my results.

Consumers know their utility from product 1 immediately:

$$u(p_1) = v_1 - p_1.$$

The only search decision is whether to visit firm 2. The expected gain from doing so is:

$$l(v_1, s, \mathbf{p}) = \int_{v_1 - p_1 + p_2}^{\bar{v}} (v_2 - p_2 - \max\{v_1 - p_1, 0\}) f(v_2) dv_2 - s. \quad (1)$$

I define the reservation value $\widehat{v}(s, \mathbf{p})$ as the solution to $l(v_1 = \widehat{v}, s, \mathbf{p}) = 0$. Intuitively, this value represents the threshold at which a consumer is indifferent between continuing to search and purchasing the current offer. It captures the opportunity cost that a consumer must forego if they decide to buy immediately rather than incur the search cost to learn about the alternative. A rational consumer computes this threshold using all available information at the time of making the search decision, including observed prices. Consequently, prices directly influence search behavior. This contrasts with the case of unposted prices, where consumers base their decision on an expectation of p_2 rather than its actual value.

Consumers buy from firm 1 without visiting firm 2 whenever product value in-hand exceeds the reservation value, $v_1 > \widehat{v}(s, \mathbf{p})$. Otherwise, they continue searching and visit firm 2. For brevity, I denote $\widehat{v}(s, \mathbf{p})$ as $\widehat{v}$ and let the price difference be $\Delta p = p_1 - p_2$.

Lemma 1 (Reservation value, $\widehat{v}$). *The reservation value – the threshold above which consumers stop searching – increases with the price of product 1, but decreases with search cost and with the price of product 2.*

To illustrate, I use $F = U[0, 1]$ as a working example throughout the paper. Under this

¹⁶Section 5 provides suggestive evidence from my empirical setting. For further evidence, see, e.g., experimental findings on position effects in sponsored search provided by (Ursu, 2018) and industry reports such as the IAB Internet Advertising Revenue Report and Google Economic Impact Report. Theoretically, assuming a free initial search is standard in sequential search models because it restores active search without imposing unnecessary frictions, while ensuring tractability in ordered search frameworks (see, e.g., Wolinsky, 1986; Arbatskaya, 2007; Zhou, 2011).

specification, the explicit form of the reservation value is:

$$\widehat{v} = 1 + \Delta p - \sqrt{2s},$$

highlighting that prices directly influence the search decision.¹⁷

Two forces shape the reservation value $\widehat{v}$. The first is the direct effect of search frictions: $\widehat{v}$ decreases with s . When search costs are high, the opportunity cost of continuing to search rises, so consumers stop searching at lower realizations of v_1 . The second force arises from *insight* prices: $\widehat{v}$ increases with the price difference $\Delta p = p_1 - p_2$. The cheaper the first firm is relative to the second, the sooner a consumer stops searching. This mechanism introduces an additional competitive dimension – firms compete not only to retain buyers but also to attract visitors – placing downward pressure on prices. This second force is central to consumer search decisions and, ultimately, to firms' pricing strategies. The formal characterization of the search rule, including conditions for continuing or stopping search, is provided by Lemma A.1.

Demand. Let D_{xy} denote the demand for firm x among consumers who visit y number of firms. For brevity, I write $D_{xy}(\mathbf{p}, \mathbf{b}, \widehat{v}(s, \mathbf{p}), \widehat{b})$ as D_{xy} . For example, D_{12} represents consumers who visit both firms but ultimately purchase from firm 1. Aggregate demand for firm 1 is $D_1 = D_{11} + D_{12}$, and for firm 2 is $D_2 = D_{22}$.

Consumers who do not continue searching ($v_1 > \widehat{v}$) purchase from firm 1:

$$D_{11} = 1 - F(\widehat{v}). \quad (2)$$

Consumers who search ($v_1 < \widehat{v}$) but prefer firm 1 after observing both valuations satisfy $v_1 - p_1 > v_2 - p_2$:

$$D_{12} = \int_{p_1}^{\widehat{v}} F(v_1 - \Delta p) \cdot f(v_1) dv_1. \quad (3)$$

Consumers who search and prefer firm 2 satisfy $v_1 - p_1 < v_2 - p_2$. This group also includes consumers who reject firm 1 ($v_1 < p_1$) and compare firm 2 to the outside option, purchasing if $v_2 > p_2$. Hence:

$$D_{22} = A \cdot [1 - F(p_2)]F(p_1) + \int_{p_1}^{\widehat{v}} [1 - F(v_1 - \Delta p)] \cdot f(v_1) dv_1, \quad (4)$$

where A denotes the “Attraction” condition from Equation A.1.¹⁸

¹⁷For comparison, under unposted prices, the expression becomes $\widehat{v} = 1 - \sqrt{2s}$.

¹⁸Consumers who reject firm 1 and compare firm 2 to the outside option face slightly different search incentives. The

3.2 Firms' Problem

In the absence of the intermediary, firms 1 and 2 set prices to maximize revenue. This exercise isolates the effect of prominence and price visibility on pricing decisions.¹⁹ The revenue functions are:

$$\max_{p_{1,NA}} Rev_1 = p_{1,NA}(D_{11} + D_{12}) \quad \text{and} \quad \max_{p_{2,NA}} Rev_2 = p_{2,NA}D_{22}, \quad (5)$$

where the subscript NA denotes the case of “No Auction.” For comparison, I use two benchmark prices: the monopoly price (p^{Mon}) and the symmetric price under random search with two firms (p^{Ran}).

3.3 Equilibrium

Using the inverse demand functions from Equations (2), (3), and (4), I simultaneously optimize the objective functions in Equation (5) to obtain equilibrium prices. Lemma 2 establishes that even when firms are symmetric in marginal cost and in the distribution of match values among consumers, there is no symmetric pure-strategy equilibrium in prices.

Lemma 2. *In the benchmark case without auction, for any positive search cost, there exists no symmetric pure-strategy equilibrium in firms' product prices.*

This result is driven by the asymmetry in demand composition arising from firms' positions. To understand the effect of this asymmetry, I solve for the asymmetric equilibrium in pure strategies. Theorem 1 characterizes the equilibrium prices.

Theorem 1 (Prices without auction). *In the benchmark case without auction, for a positive search cost, there is a unique asymmetric pure-strategy equilibrium in firms' product prices. The equilibrium prices satisfy the following implicit conditions, where A denotes the Attraction condition (see appendix for details).*

$$p_{1,NA}^* = \frac{1 - F(\widehat{v}^*) + \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f(v_1) dv_1}{f(\widehat{v}^*) - \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f'(v_1) dv_1},$$

$$p_{2,NA}^* = \frac{A \cdot [1 - F(p_{2,NA}^*)]F(p_{1,NA}^*) + F(\widehat{v}^*) - F(p_{1,NA}^*) - \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f(v_1) dv_1}{A \cdot f(p_{2,NA}^*)F(p_{1,NA}^*) + f(\widehat{v}^*) - F(p_{2,NA}^*)f(p_{1,NA}^*) - \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f'(v_1) dv_1}.$$

The implicit expressions in Theorem 1 arise because each firm's optimal price depends on demand elasticities that are themselves functions of reservation value, which varies with both prices and search cost. For firm 1, the numerator captures marginal revenue from consumers

formal derivation of the “Attraction” condition and its implications is provided in Appendix A.2.2.

¹⁹The objective functions can also be interpreted as profit functions in settings where firms pay a fixed cost for prominence, such as offline advertising. See [Hristakeva and Mortimer \(2021\)](#) for an overview of television advertising, a major component of pre-digital advertising expenditure.

who buy immediately and those who return after searching, while the denominator reflects the sensitivity of this demand to price changes. For firm 2, the numerator includes demand from consumers who search and those attracted under the “Attraction” condition, whereas the denominator accounts for the higher elasticity caused by its dual role of attracting visitors and converting them into buyers. This interdependence between prices and search behavior prevents closed-form solutions, leading to implicit equilibrium conditions.

Although closed-form solutions are not available, the structure of these expressions allows us to infer that $p_1^* > p_2^*$ under standard regularity conditions. The numerator for p_1^* includes $(1 - F(\widehat{v}^*))$ and adds an integral term capturing substitution effects, while the numerator for p_2^* subtracts this same integral and depends on $F(p_1^*)$, which decreases as p_1^* rises. Similarly, the denominator for p_2^* contains additional positive terms such as $Af(p_2^*)F(p_1^*)$, making it larger than the denominator for p_1^* . A smaller numerator combined with a larger denominator implies that p_2^* is lower than p_1^* . Under log-concavity of F and monotonic hazard rates, these conditions hold, and for the Uniform case, the result is immediate since higher-order derivatives vanish and the simplified expressions confirm $p_1^* > p_2^*$.

The prominent firm (position 1) faces less elastic demand because consumers start their search there and may stop after observing its price. In contrast, the non-prominent firm competes for consumers who are willing to incur an additional search cost, making its demand more elastic. With observable or posted prices, firm 2 must set a lower price to attract visitors, while firm 1 can maintain a higher price without losing much demand. This asymmetry in elasticity explains why p_1^* exceeds p_2^* in equilibrium.

The intuition behind the price expressions in Theorem 1 is further illustrated by the working example of $F = U[0, 1]$ as follows. The formal characterization of the results behind Figures 1 and 2 from this case is provided by Proposition 1 and Corollary 1.

Proposition 1. *When match values are drawn from a Uniform distribution ($v \sim F = U[0, 1]$), the reservation value is given by $\widehat{v} = 1 + \Delta p - \sqrt{2s}$ and*

- *the search cost at which the “Attraction” condition starts binding is given by $\underline{s}_{NA} = \frac{1}{4}$, and the search cost at which firm 2 drops out of the market is given by $\bar{s}_{NA} = \frac{1}{2}$,*
- *the price of prominent firm increases with search cost $\left(\frac{dp_{1,NA}^*}{ds} > 0\right)$, and the price of non-prominent firm decreases with search cost $\left(\frac{dp_{2,NA}^*}{ds} < 0\right)$,*
- *and the price difference is positive: $p_{1,NA}^* > p_{2,NA}^*$*

Corollary 1. *The price of the prominent firm lies between p^{Ran} (at $s = 0$) and monopoly price (at*

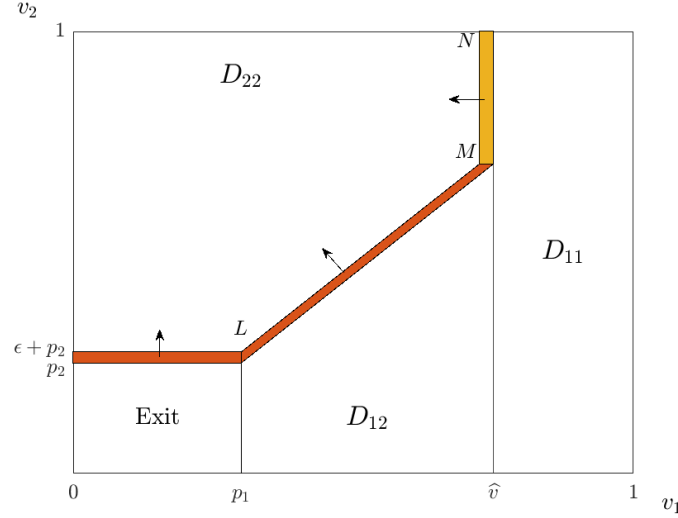


Figure 1: Elasticity of demand under posted prices: Non-prominent firm faces more elastic demand, explaining why its price falls relative to the prominent firm as search cost declines

$s = \bar{s}$), while the price of the non-prominent firm lies between p^{Ran} (at $s = 0$) and 0 (at $s = \bar{s}$).

$$p_{1,NA}^* \in [p^{Ran}, p^{Mon}] \quad , \quad p_{2,NA}^* \in [0, p^{Ran}]$$

Consider the unit mass of consumers represented in Figure 1, where each point (v_1, v_2) in the square corresponds to a consumer type with those match values. This plot summarizes the role of consumer heterogeneity and illustrates each firm's objective.

For a small increase in p_2 , firm 2's demand changes along two segments: LM and MN . Segment LM represents consumers who visit both firms; changes in p_2 affect their choice between firms, and since firm 1 also competes for these consumers, its demand responds symmetrically to changes in p_1 . The asymmetry arises from segment MN , which represents consumers who decide not to visit firm 2 because of the higher p_2 . Firm 1 does not face this trade-off, placing firm 2 at a competitive disadvantage.

In summary, firm 2's price serves a dual role: attracting visitors and converting them into buyers. This dual role makes its demand more elastic and explains why equilibrium prices are asymmetric.²⁰

Figure 2 summarizes the effect of search costs on pricing. At zero search cost, the model collapses to the case of [Perloff and Salop \(1985\)](#), yielding the full-information price as the

²⁰To further understand the effect of posted prices, I disentangle the channels influencing firm 2's decision in Online Appendix Section A, which also compares posted and unposted prices.

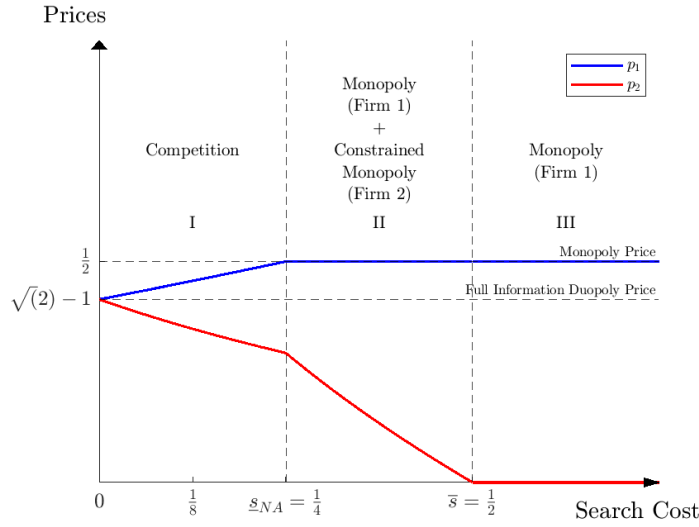


Figure 2: Equilibrium prices without auction: Prominent firm charges a higher price ($p_1 > p_2$), and the gap widens with search cost, contrasting with standard prominence models

equilibrium. As search cost increases, fewer consumers search. Competition therefore declines gradually with s in region I, as seen from the increase in firm 1's price. One might expect firm 2's price to rise as well. However, because prices are posted, firm 2 cannot profitably deviate upward: doing so would immediately reduce its visitor base. For firm 2, this loss outweighs the benefit of charging a higher price to the remaining "high-interest" visitors. Although the fraction of high-interest visitors rises with search friction, the loss of other visitors also grows. Consequently, even as competition weakens, p_2 decreases with s in equilibrium.

At intermediate search costs ($\underline{s} < s < \bar{s}$), the fraction of high-interest visitors reaches its limit. A segment of consumers buys product 1 without considering the alternative, while the rest find it unaffordable at any price. Firm 2's demand now consists only of consumers whose outside option yields zero utility, meaning they will not return to firm 1 regardless of its offer. These consumers either buy product 2 or exit the market. In effect, competition disappears, and the market becomes segmented: no consumer compares both products when purchasing. Both firms would prefer to set monopoly prices, but since the "Attraction" condition (Equation A.1) binds for $s > \underline{s}$, firm 2 cannot raise its price without losing all visitors. It therefore chooses a constrained-optimal price to attract at least some buyers, ensuring positive revenue and continued participation.

At high search costs ($s > \bar{s}$), firm 2 exhausts its ability to attract consumers despite search frictions. Its price falls to zero, leaving firm 1 as the sole active firm, effectively operating as a

monopoly.²¹

3.4 Taking Stock

This section analyzed the benchmark case without auction to isolate the role of search frictions and prominence in shaping pricing strategies. The key takeaways are as follows. Even with symmetric firms, when prices are observable prior to search, prominence creates an inherent asymmetry in demand composition, leading to an asymmetric pure-strategy equilibrium in prices. Firms compete not only to retain buyers but also to attract visitors, introducing an additional competitive dimension absent in models with hidden prices.

These observations establish a benchmark for analyzing how search frictions and price observability influence competitive outcomes in the absence of an intermediary. By clarifying how prominence and search frictions interact with price visibility in a simplified setting, this section sets the stage for the mechanisms examined in Section 4, where position auctions and CPC payments amplify these forces to determine market outcomes.

4 Market Equilibrium With Auction

In this section, I introduce a revenue-maximizing intermediary that conducts an auction to (i) allocate positions to ads and (ii) determine cost per click (CPC).²²

4.1 Position Auction

Firms' Problem. Firms participate in a one-shot auction conducted by the intermediary to secure an ad position. After submitting bids (their willingness to pay per click), firms learn their positions and corresponding CPC. Finally, they set product prices. Formally, each firm maximizes profit by choosing its auction bid (b_1, b_2) and product price (p_1, p_2) :

$$\max_{p_1, b_1} p_1 \cdot D_1(p_1, p_2, s) - r_1(b_i, b_j, \widehat{b}) \cdot \text{Clicks}_1(p_1, p_2, s), \quad (6)$$

$$\max_{p_2, b_2} p_2 \cdot D_2(p_1, p_2, s) - r_2(b_i, b_j, \widehat{b}) \cdot \text{Clicks}_2(p_1, p_2, s). \quad (7)$$

Intermediary's Problem. The intermediary conducts a one-shot Generalized Second Price (GSP) auction with the objective of maximizing its own revenue. Firms are ranked based on expected payment, defined as the CPC multiplied by the expected number of clicks. The intermediary can choose between a monopoly and duopoly product market; ties in the duopoly

²¹Note that $s = \frac{1}{2}$ is also the search cost at which market breakdown occurs in Wolinsky (1986) and Armstrong, Vickers, and Zhou (2009).

²²In an intermediate step, Online Appendix Section C characterizes equilibrium product prices under exogenous CPC.

case are broken by displaying each possible order with equal probability.

The intermediary sets the optimal reserve price $\widehat{b}$ to maximize revenue:

$$\max_{\widehat{b}} \pi_m = r_1(b_i, b_j, \widehat{b}) \cdot \text{Clicks}_1(p_1, p_2, s) + r_2(b_i, b_j, \widehat{b}) \cdot \text{Clicks}_2(p_1, p_2, s).$$

The reserve price $\widehat{b}$ is announced before the auction and is common knowledge. In a GSP auction, firms pay a per-click commission equal to the bid of the next firm in the ranking: $r_1 = b_2$ and $r_2 = \widehat{b}$.²³

4.2 Equilibrium

Given consumer demand and firm revenues as implicit functions of prices for each position (see Section 3), I now solve for the optimal bidding behavior of the two firms. Once optimal bids are determined, I derive the optimal auction reserve price. When firms bid symmetrically:

$$b_i = b_j = Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})),$$

which is feasible only if:

$$\widehat{b} \leq \frac{Rev_2}{1 - D_{11}}.$$

When firms bid asymmetrically:

$$\begin{aligned} b_i &\in (Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})), \infty), \\ b_j &\in [\widehat{b}, Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11}))], \end{aligned}$$

which is feasible only if:

$$\widehat{b} \leq \min \left\{ \frac{Rev_1 - Rev_2}{D_{11}}, \frac{Rev_2}{1 - D_{11}} \right\}.$$

Appendix Table A.1 summarizes market outcomes for different values of the auction reserve price $\widehat{b}$. The condition for the optimal reserve price is derived from firms' participation and incentive compatibility constraints. As shown in the literature, this specification can lead to multiple equilibria in pure strategies. Similar to the refinement used in second-price auctions, I focus on equilibria in undominated strategies, which rule out asymmetric bidding and yield

²³See Online Appendix Sections E, F, and H for a discussion on timing and on auction format.

a unique equilibrium outcome for a given search cost.²⁴ This outcome coincides with the refinement of “locally envy-free” strategies proposed by [Edelman, Ostrovsky, and Schwarz \(2007\)](#). Empirical evidence in Section 5 supports the view that firms’ bids are strategic complements, consistent with this refinement.

Theorem 2 (Equilibrium with Auction). *In the full equilibrium with auction, optimal bids and the auction reserve price in the GSP auction are:*

$$\begin{aligned}\widehat{b}^* &= \min \left\{ Rev_1^* - Rev_2^*, \frac{Rev_2^*}{F(\widehat{v}^*)} \right\}, \\ b_i^* &= b_j^* \in (Rev_1^*, \infty).\end{aligned}\tag{8}$$

Firms occupy each position with probability one-half. Equilibrium prices in pure strategies for each position are:

$$\begin{aligned}p_1^* &= \frac{1 - F(\widehat{v}^*) + \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p) f(v_1) dv_1}{f(\widehat{v}^*) - \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p) f'(v_1) dv_1}, \\ p_2^* &= \frac{A \cdot [1 - F(p^*)]F(p^*) + F(\widehat{v}^*) - F(p_1^*) - \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p) f(v_1) dv_1 + \widehat{b}^* f(\widehat{v}^*)}{A \cdot f(p^*)F(p^*) + f(\widehat{v}^*) - F(p_2^*)f(p_1^*) - \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p) f'(v_1) dv_1}.\end{aligned}\tag{9}$$

Comparison with Theorem 1. Similar to the benchmark case without auction (Theorem 1), equilibrium prices under auction are obtained by simultaneously optimizing firms’ objectives. However, the introduction of the intermediary and CPC payments fundamentally changes the pricing problem. In Theorem 1, asymmetry in prices arose solely from differences in demand composition due to prominence. In contrast, Theorem 2 shows that the auction introduces an additional asymmetry through CPC: the non-prominent firm faces a marginal cost-like term in its pricing condition, while the prominent firm does not.

The expressions in Theorem 2 indicate that the introduction of CPC payments alters the relative magnitude of prices compared to Theorem 1. For p_2^* , the numerator now includes an additional term $\widehat{b}^* f(\widehat{v}^*)$, which acts like a marginal cost component, while p_1^* remains unaffected by CPC. This extra term reduces firm 2’s residual demand and increases its effective marginal cost, making its pricing condition more restrictive. This pushes p_2^* upward relative to the no auction case, while p_1^* rises only slightly because its demand remains relatively insensitive to CPC. For sufficiently low search costs, the reversal $p_2^* > p_1^*$ can occur, as firm 2’s marginal cost effect dominates its elasticity advantage.

Intuitively, the auction creates a two-stage game. Firms first compete for positions by bidding for clicks, and then set product prices conditional on their position and CPC payment. CPC acts

²⁴This refinement assumes bidders exercise caution and do not completely rule out deviations by rivals. See, for instance, [Brandenburger, Friedenberg, and Keisler \(2008\)](#) and [Hillas and Samet \(2020\)](#).

as a pass-through channel: for the prominent firm, it behaves like a fixed cost because its click volume is relatively insensitive to its price; for the non-prominent firm, it acts as a marginal cost because its click volume is more sensitive to its price. This dual role of price for the non-prominent firm – attracting visitors *and* converting them into buyers – makes its demand more elastic and constrains its ability to raise prices. By contrast, the prominent firm can increase its price without losing its visitor base, reinforcing its competitive advantage.

The intuition behind the effect of the GSP auction on firm outcomes in Theorem 2 is further illustrated by the working example of $F = U[0, 1]$ as follows. The formal characterization of the results behind Figure 3 from this case is provided by Proposition 2 and Corollary 2.

Proposition 2. *When match values are drawn from a Uniform distribution ($v \sim F = U[0, 1]$),*

- *the intermediary sets a reserve price to extract all revenue from the non-prominent firm, $\widehat{b} = \frac{Rev_2}{1-D_{11}}$,*
- *firms bids are symmetric and they compete aggressively to end up transferring all the benefits of prominence to the intermediary, $b_i = b_j = Rev_1$.*
- *Hence, firm profits are zero, $\pi_1 = \pi_2 = 0$,*
- *and intermediary revenue equals total firm revenue, $\pi_m = Rev_1 + Rev_2$*

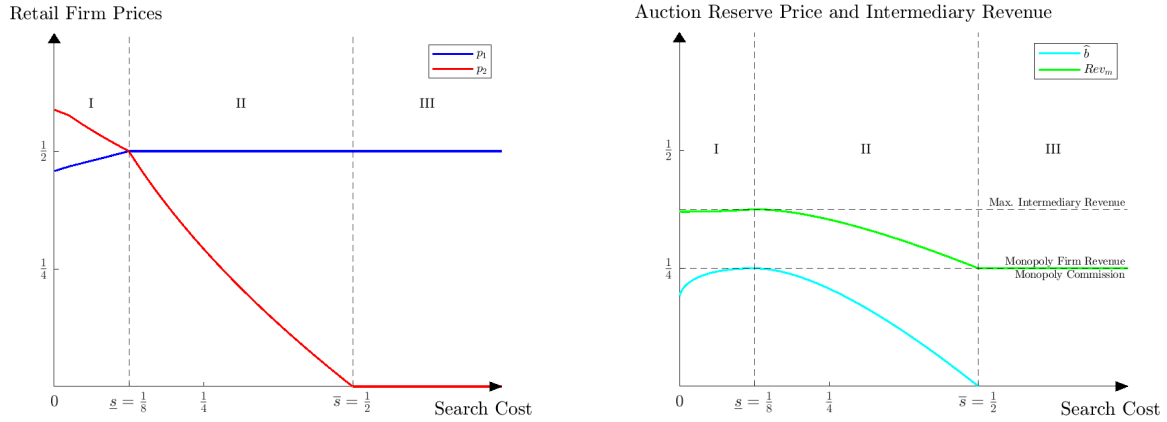
Corollary 2. *Comparison with ‘No-Auction’ case ($r_1 = r_2 = 0$):*

$$p_1 \geq p_{1,NA} \text{ and } p_2 \geq p_{2,NA} \text{ s.t. } \underline{s} = \frac{1-2\widehat{b}^*}{4} < (\underline{s})_{NA}$$

Figure 3a illustrates how the auction mechanism reshapes market outcomes compared to the benchmark case without auction. First, both equilibrium prices increase relative to the No-Auction case, reflecting the additional CPC component in firms’ pricing decisions. This upward shift means that prices reach monopoly levels at a lower search cost, accelerating the loss of competitive pressure.

Second, the reservation value $\widehat{v}$ declines more steeply with search cost under auction. This is driven by the relatively larger increase in p_2 , which reduces the expected gains from continued search and discourages consumers from visiting the non-prominent firm. As a result, Region I (the zone of active competition) shrinks significantly, and market segmentation occurs at a lower threshold of search cost compared to the benchmark case.

Third, the figure highlights the asymmetric impact of CPC on firms. While both firms raise prices, the non-prominent firm faces a stronger constraint: its price includes a marginal CPC term, which amplifies its sensitivity to search frictions. This explains why p_2 rises sharply initially but then flattens as firm 2 approaches its participation constraint. In contrast, firm 1 benefits from its position advantage and the fixed-cost nature of CPC, allowing it to maintain higher prices without losing demand.



(a) Prices: Both firms raise prices with search cost, but the non-prominent firm faces stronger CPC-induced constraints, accelerating segmentation.

(b) Optimal CPC and intermediary revenue: Revenue and CPC rise initially with search cost but decline once segmentation reduces click volume.

Figure 3: Equilibrium outcomes with auction

Overall, the figure demonstrates that the auction mechanism not only increases prices but also compresses the competitive region and accelerates market segmentation. These dynamics underscore the dual role of CPC as both a bidding cost and a pricing constraint, reinforcing the prominent firm's advantage and weakening the non-prominent firm's ability to compete.

Figure 3b illustrates how intermediary revenue and the optimal CPC evolve with search cost. Initially, as search cost rises from zero, intermediary revenue increases. This occurs because higher search frictions reduce consumer switching, allowing firms to bid more aggressively for prominent positions. The platform benefits from this intensified bidding competition, which raises CPC and, in turn, its revenue.

However, beyond a certain threshold of search cost, intermediary revenue begins to decline. At high search costs, market segmentation becomes pronounced: consumers increasingly settle for the first option and avoid searching altogether. This reduces the volume of clicks, which are the basis for CPC payments. Even though CPC remains positive, the sharp fall in click volume dominates, causing overall revenue to drop.

The optimal CPC follows a similar pattern. It rises with search cost in the initial region, reflecting the platform's incentive to exploit competitive bidding when both firms actively seek visitors. Once segmentation sets in and firm 2's participation becomes constrained, the optimal CPC declines because raising it further would deter bidding and reduce click-based revenue.

In summary, Figure 3b highlights a non-monotonic relationship: intermediary revenue and optimal CPC increase with search cost when competition is active but decrease once search

frictions erode consumer search and click volume. This dynamic underscores the trade-off faced by the platform between extracting higher CPC payments and maintaining sufficient consumer engagement.

4.3 Welfare

Having characterized equilibrium prices and bidding behavior, I now turn to welfare implications. Figures A.2 (without auction) and A.3 (with auction) plot Consumer Surplus (CS), Industry Revenue (IR), and Total Welfare (TW) for the working example of $F = U[0, 1]$.²⁵

Consumer Surplus. Compared to the No-Auction case, CS is lower with auction because prices are higher under the auction mechanism. Market segmentation occurs at a lower search cost ($\underline{s} < \underline{s}_{NA}$), shifting the welfare curves to the left.

Industry Revenue. Under the No-Auction benchmark, IR reflects only firms' product revenues and rises with search cost as reduced consumer search leads to more inelastic consumers, which allows higher prices. In contrast, under auction, IR includes both product revenues and intermediary revenue from CPC payments. This addition amplifies the initial increase in IR because bidding competition intensifies as search frictions grow. However, the decline in IR at high search costs is sharper under auction: segmentation reduces not only transaction volume but also click-based payments. Thus, while the auction mechanism boosts IR at low to moderate search costs, it accelerates its collapse once consumer engagement falls.

Total Welfare. TW peaks at a lower search cost than IR , highlighting a divergence between platform incentives and overall efficiency. While the platform's revenue-maximizing choice of search cost occurs near the onset of segmentation, the welfare-maximizing point is reached earlier, when competition still benefits consumers. These dynamics underscore the potential tension between intermediary design choices and consumer welfare.

Extensions. In Appendix Section C.1, I compare alternative platform revenue models – cost per click (CPC), cost per purchase (CPP), and cost per view (CPV) – and find that the choice of bidding format significantly alters pricing incentives and welfare outcomes. Under CPC, the intermediary's CPC charge introduces a marginal cost component for the non-prominent firm, amplifying asymmetry in pass-through and accelerating segmentation. In contrast, CPP aligns bidding costs with successful transactions, reducing the distortionary effect on search incentives and leading to more balanced pricing. CPV, which charges for exposure rather than engagement, weakens the link between bidding and conversion, resulting in the smallest impact on product prices but the largest potential for inefficient allocation of prominence. These comparisons underscore that platform design is not neutral: revenue models shape the intensity

²⁵For detailed welfare analysis under the No-Auction benchmark, including conversion rates and transaction volumes, see Appendix A.2.4.

of competition, the distribution of surplus, and the degree of market segmentation.

4.4 Taking Stock

This section has shown how the introduction of a position auction and CPC payments reshapes market outcomes. Compared to the benchmark case without auction, prices rise, the competitive region shrinks, and market segmentation occurs earlier. The auction amplifies asymmetry between firms: the prominent firm benefits from CPC acting as a fixed cost, while the non-prominent firm faces CPC as a marginal cost, constraining its pricing flexibility. These dynamics reinforce the prominent firm's advantage and accelerate the erosion of competition.

Empirical Prediction. The intuitive takeaway from this section is that Intermediary revenue is maximized at the lowest search cost where competition disappears and each product serves an exclusive segment. If the platform can influence search cost (e.g., through interface design), it would choose $\underline{s}$. At this point, conversion rates across positions converge, eliminating asymmetry in pass-through. This provides a testable implication: when conversion rates are equal, symmetric firms should charge similar prices. Section 5 presents empirical evidence in support of this prediction.

Extensions. The analysis also extends to the case of asymmetric firms (see Appendix Section C.2), where vertical differentiation in products amplifies the effects documented above. Results are robust and, in fact, stronger: asymmetry in products interacts with prominence and CPC payments to widen the pricing gap and accelerate market segmentation. These findings underscore that the mechanisms identified in the symmetric benchmark persist under more general conditions and become even more pronounced.

5 Data and Evidence

This section examines whether the patterns predicted by the model are reflected in the data. The analysis focuses on two key outcomes: (i) variation in cost-per-click (CPC) across positions and its link to pricing asymmetry, and (ii) convergence in conversion rates as a proxy for symmetric pass-through. These patterns provide a testable implication of the theoretical mechanism: when conversion rates across positions equalize, price differences should vanish. While the evidence is descriptive and does not establish causality, it offers suggestive yet novel support for the model's predictions on conversion rates. The rest of this section proceeds as follows: First, I discuss the strengths and limitations of the dataset. Second, I summarise and describe the patterns in the data. Finally, I present evidence in support of the hypotheses.

5.1 Data and Context

The dataset (Diemert Eustache, Meynet Julien, Galland, and Lefortier, 2017) comprises 30 days of user traffic and the ads displayed on a large anonymous platform in the United States. It contains detailed information on both ad-related payments and user behavior, offering a unique opportunity to examine interactions between consumers and firms in a sponsored advertising environment.²⁶ A typical user session involves (i) viewing ordered ads, (ii) clicking to learn product details, and (iii) potentially making a purchase. Each observation corresponds to an ad impression shown to a user. The data has been sub-sampled and anonymized to protect proprietary elements.²⁷ The dataset includes a unique identifier for each user and each ad campaign,²⁸ as well as indicators for whether the ad was clicked or purchased, timestamps for clicks and purchases, the position at which the ad was displayed, the position of the click in the user’s search order, whether the eventual purchase was attributed to the platform, the CPC paid to the intermediary for the click, and the cost-per-order (CPO) borne by the firm.

Two caveats merit attention before proceeding. First, the dataset does not specify the auction format used. Since sponsored ads are typically allocated via a per-click second-price auction, I adopt this assumption.²⁹ Consistent with the theoretical model, I abstract from dynamic strategic considerations in bidding. Second, as with most position-auction datasets, posted product prices are not observed. However, the dataset reports CPO, which I use as a proxy for product price. In standard industry practice, CPO includes customer acquisition, packaging, fulfillment, shipping, cost of goods sold, and storage costs, averaged per order. All subsequent analysis is subject to these assumptions.

5.2 Descriptive Statistics

Table 1 summarizes the dataset, which covers 30 days of user traffic and ad impressions on a large platform. I restrict the sample to observations where exactly two ads were displayed, allowing a clean identification of competitors and alignment with the theoretical setting. Purchases attributed to the platform are retained to avoid concerns of showrooming.

A key feature of user behavior is that the order of clicks always follows the order in which

²⁶To my knowledge, this dataset has not previously been used for economic or marketing analysis. It was originally compiled by a large digital marketing agency to train and evaluate machine learning models for performance optimization. See Appendix D for additional details.

²⁷The organization has not formally disclosed whether the sampling was random.

²⁸In digital marketing terminology, a “campaign” refers to an advertising unit created by a firm (e.g., Google Ads campaigns, see [Google Ads Help](#)). Google Ads While the identifier could represent a retailer-product pair, it is unlikely that a firm would run two distinct campaigns for the same product on the same platform within 30 days. Accordingly, I assume each campaign corresponds to a single product. The implications of a multi-product firm are beyond the scope of this paper

²⁹See, for example, <https://support.google.com/adsense/answer/160525?hl=en>.

Table 1: Data description and click probabilities by position

Number of users	985,273
Number of users who made at least one click	480,536
Number of users who made at least one purchase	42,673
Number of ads	675
Total observations	1,612,594
$\mathbb{P}[\text{click if prominent}]$	0.36
$\mathbb{P}[\text{click if non-prominent}]$	0.10
$\mathbb{P}[\text{click if non-prominent} \mid \text{prominent clicked}]$	0.27
$\mathbb{P}[\text{click if non-prominent} \mid \text{prominent not clicked}]$	0.00

Notes: The table summarizes user activity and click probabilities by ad position. Probabilities are unconditional unless stated otherwise. The data confirm ordered search: users rarely click the second ad without first clicking the prominent ad.

ads are displayed, consistent with the assumption of ordered search in the model. Additional descriptive patterns, including click-through rates, conversion behavior, and distributional comparisons of key variables across positions, are provided in Appendix D. These details confirm the prominence effect and motivate the empirical strategy.

5.3 Empirical Analysis

To examine whether the patterns predicted by the model appear in the data, I estimate:

$$y_{ijt} = \alpha_j + \alpha_t + \beta \cdot \text{prominence}_{ijt} + \epsilon_{ijt}, \quad (10)$$

for user i facing ad j at time t . The outcome y includes (i) cost per click (CPC), (ii) click indicator, (iii) conversion indicator conditional on click, and (iv) cost-per-order (proxy for price). The explanatory variable *prominence* equals one if the ad is displayed in the top position. Ad fixed effects (α_j) control for unobserved ad characteristics, and time fixed effects (α_t) account for seasonality. Standard errors are clustered at the ad level. Estimates are correlational rather than causal, but are informative about systematic differences between the same ad shown in different positions.

Table 2 reports the results. Ads in the prominent position pay higher CPC and receive more clicks, consistent with the mechanism of asymmetric pass-through. Conversion rates and proxy prices do not differ significantly, which aligns with the prediction that prominence affects bidding costs rather than intrinsic product valuation. These findings support the model’s implication that CPC introduces a marginal cost component for the non-prominent firm, amplifying pricing asymmetry.³⁰

³⁰Appendix D provides additional heterogeneity analysis and tests for strategic complementarity in bids following

Table 2: Effect of prominence on CPC, clicks, conversion, and price

Outcome:	(1) CPC	(2) clicks	(3) conversion rate	(4) cost-per-order	(5) first click
prominence	0.0746*** (0.021)	0.00808*** (0.002)	-0.00130 (0.001)	-0.0389 (0.026)	1*** (0.000)
Observations	1,594,107	1,594,107	460,785	1,594,107	460,785
R-squared	0.050	0.029	0.142	0.823	1.000
mean(y)	1.921	0.297	0.0781	20.80	
sd(y)	5.395	0.457	0.268	12.07	
ad FE	✓	✓	✓	✓	✓
day of the week FE	✓	✓	✓	✓	✓
hour of the day FE	✓	✓	✓	✓	✓

Notes: Each column reports estimates from Equation (10), where the outcome is indicated in the header. The coefficient on *prominence* captures the difference between the same ad displayed in the top position versus the lower position. Results show that prominent ads pay higher CPC and receive more clicks, while conversion rates and cost-per-order (proxy for price) remain similar. Standard errors clustered at the ad level. All specifications include ad fixed effects and time controls.

6 Conclusion and Discussion

The two-sided nature of online marketplaces makes intermediaries central to shaping economic outcomes. This paper develops a model of consumer-firm and firm-intermediary interaction to illustrate the trade-offs faced by each player. I consider an ordered search framework where consumers discover product valuations sequentially by incurring search costs, while the intermediary allocates positions through an auction. Price observability prior to search plays a critical role in determining consumer demand and firm profits. Even when prices are fully visible and the intermediary does not compete directly with firms, the interaction between search frictions and auction design can exacerbate market concentration.

The analysis shows that adopting price visibility can improve consumer surplus: observable or posted prices intensify competition by forcing the non-prominent firm to lower its price, preventing surplus extraction from held-up consumers. This intervention is straightforward to implement through minor platform design changes. However, caution is warranted – while price visibility reduces hold-up, it also discourages search and may create entry barriers in the long run. Evidence on dynamic effects is needed to fully assess welfare implications. Another policy lever is adopting a *cost-per-purchase* (CPP) model, albeit at the expense of intermediary revenue.

The model generates predictions for two empirically relevant metrics: conversion rates and

De Paula and Tang (2012), which support the equilibrium refinement used in Section 4.

transaction volumes. Conversion rates provide a practical benchmark for evaluating the pass-through of advertising costs into product prices. For firms, higher conversion rates signal effective advertising investment; for policymakers, they capture the welfare impact of CPC charges acting as marginal costs. When CPC burdens the non-prominent firm, its price rises, reducing conversion rates and overall transactions, while enabling the prominent firm to raise its price without losing demand. This asymmetric dynamic offers a novel rationale for growing market concentration.

Sponsored ads are often viewed as informative, guiding consumers toward relevant products. Recent work, however, emphasizes their persuasive potential, particularly when intermediaries steer consumers toward preferred ads (Heidhues and Kőszegi, 2018). My analysis shows that such steering can arise endogenously: the intermediary-preferred product may secure prominence and set a lower price, persuading rational consumers without explicit nudging. Extensions to asymmetric firms indicate that advertising can act as a coordination device, placing higher-value firms in prominent positions and reinforcing their advantage, echoing insights from Bagwell and Ramey (1994) and Coles and Eeckhout (2000). While the analysis is framed in the context of digital advertising, the underlying mechanism – prominence combined with position-dependent commissions and observable prices – applies more broadly to settings where sellers pay for display priority, such as shelf placement in retail, promoted listings on e-commerce platforms, and commission-based financial product distribution.

The topic of digital marketplaces deserves further research theoretically and empirically. Prominence may affect entry incentives: while high search costs and fixed CPC charges can deter entry and reduce variety (Lee and Musolf, 2021), they may also mitigate inefficiencies from excess entry under random search (Anderson and Renault, 1999). More broadly, the asymmetry in CPC cost structure – fixed for the prominent firm, marginal for the non-prominent firm – suggests a mechanism through which auction-based platforms may widen the gap between established and fringe firms. Since firm revenues are distributed in order of position and this dispersion grows with search cost, a revenue-maximising intermediary may endogenously reinforce the advantage of large incumbents, providing a rationale for the growing market concentration documented in recent empirical work (De Loecker, Eeckhout, and Unger, 2020). Empirical estimation of CPC pass-through remains an open challenge and could shed light on the strength of these mechanisms. Welfare analyses incorporating heterogeneity and long-run dynamics, such as endogenous consumer loyalty, are essential for understanding the persistence of market power and concentration in digital marketplaces.

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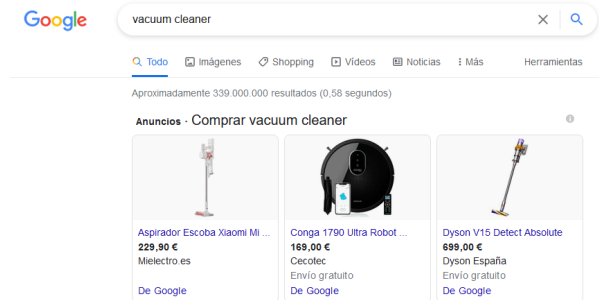
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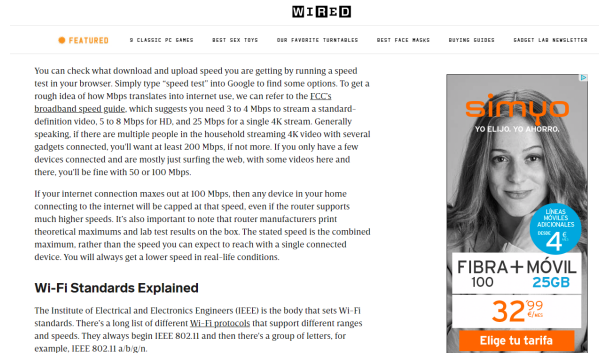
A Appendix: Theoretical Analysis

This appendix provides supplementary figures and tables referenced in the main text, followed by derivations and proofs.

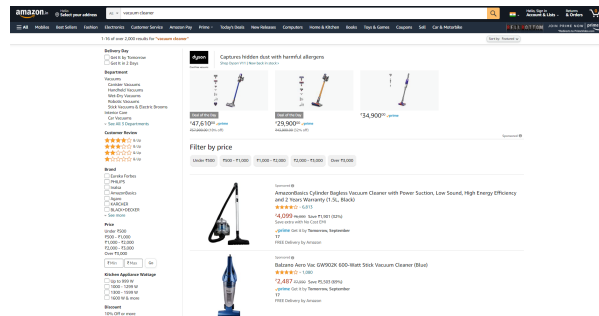
A.1 Figures and Tables



Source: <https://www.google.com/> (Accessed: 16 Sep, 2021)

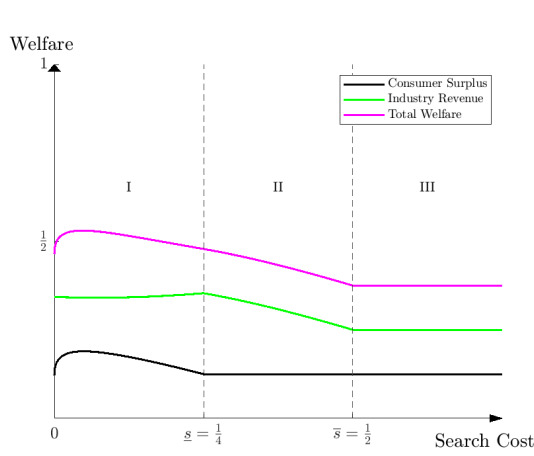


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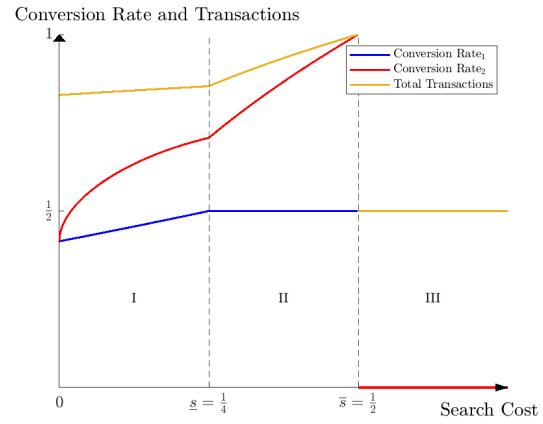


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Figure A.1: Examples of sponsored results on digital platforms: Prices are prominently displayed and observable at zero cost, while product characteristics require additional search effort.

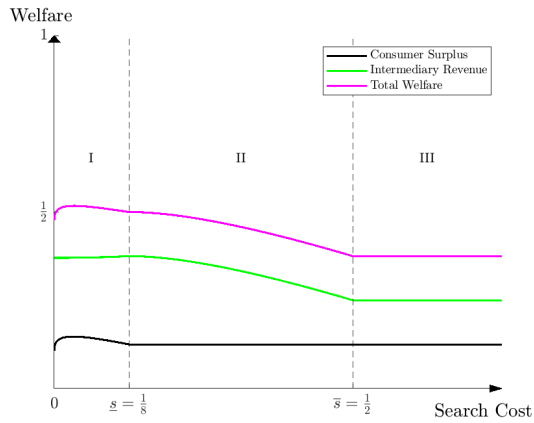


(a) Welfare components: Consumer surplus declines with search cost, while total welfare peaks earlier than industry revenue.

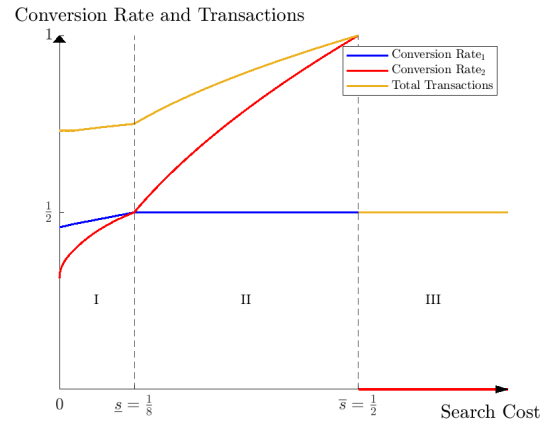


(b) Conversion rate: Prominent firm's conversion rate rises with search cost, while the non-prominent firm's falls, reflecting asymmetric elasticity.

Figure A.2: Welfare and efficiency under the benchmark case without auction.



(a) Welfare components: Consumer surplus declines with search cost, while total welfare peaks earlier than intermediary revenue.



(b) Conversion rates: Prominent firm's conversion rate rises with search cost, while the non-prominent firm's falls, reflecting asymmetric elasticity.

Figure A.3: Welfare and efficiency under auction.

Table A.1: Equilibrium outcomes for different auction reserve prices

Auction Reserve Price	Firms find ...	Firms' Bids	Firms' Profits
$\widehat{b} > Rev^{Mon}$	neither feasible*	$b_1 = \times$ $b_2 = \times$	$\pi_1 = 0$ $\pi_2 = 0$ $Rev_m = 0$
$\max \left\{ \frac{Rev_1 - Rev_2}{D_{11}}, \frac{Rev_2}{1 - D_{11}} \right\} < \widehat{b} < Rev^{Mon}$	position 1 feasible	$b_1 = Rev^{Mon}$ $b_2 = Rev^{Mon}$	$\pi_1 = 0$ $\pi_2 = 0$ $Rev_m = Rev^{Mon}$
$\frac{Rev_2}{1 - D_{11}} < \widehat{b} < \frac{Rev_1 - Rev_2}{D_{11}}$	position 1 feasible	$b_1 = Rev^{Mon}$ $b_2 = Rev^{Mon}$	$\pi_1 = 0$ $\pi_2 = 0$ $Rev_m = Rev^{Mon}$
$\max \left\{ \frac{Rev_2}{2}, \frac{Rev_1 - Rev_2}{D_{11}} \right\} < \widehat{b} < \frac{Rev_2}{1 - D_{11}}$	neither feasible	$b_1 = \times$ $b_2 = \times$	$\pi_1 = 0$ $\pi_2 = 0$ $Rev_m = 0$
$\frac{Rev_1 - Rev_2}{D_{11}} < \widehat{b} < \frac{Rev_2}{2} < \frac{Rev_2}{1 - D_{11}}$ (this interval has zero measure for $U[0, 1]$)	position 2 feasible	$b_1 = \widehat{b}$ $b_2 = \widehat{b}$	$\pi_1 = -\widehat{b}$ $\pi_2 = \min \left\{ \frac{1}{2}, 1 - \sqrt{2s} \right\} - \widehat{b}$ $Rev_m = \widehat{b} \leq \frac{Rev^{Mon}}{2}$
$\widehat{b} \in \left[0, \frac{Rev_2}{1 - D_{11}} \right]$	both feasible <i>symmetric bids</i>	$b_1 = Rev_1 - \pi_2$ $b_2 = Rev_1 - \pi_2$	$\pi_1 = Rev_1 - b_2$ $\pi_2 = Rev_2 - \widehat{b}(1 - D_{11})$ $Rev_m = \frac{b_2 + \widehat{b}(1 - D_{11})}{2}$
$\widehat{b} \in \left[0, \min \left\{ \frac{Rev_1 - Rev_2}{D_{11}}, \frac{Rev_2}{1 - D_{11}} \right\} \right]$	both feasible <i>asymmetric bids</i>	$b_1 \in (Rev_1 - \pi_2, \infty)$ $b_2 \in [\widehat{b}, Rev_1 - \pi_2]$	$\pi_1 = Rev_1 - b_2$ $\pi_2 = Rev_2 - \widehat{b}(1 - D_{11})$ $Rev_m = b_2 + \widehat{b}(1 - D_{11})$

Notes: The table summarizes equilibrium outcomes for varying values of the auction reserve price $\widehat{b}$, listed in descending order. Feasibility refers to whether firms earn non-negative profits. Rev^{Mon} denotes monopoly revenue; Rev_k is revenue of firm in position k . Outcomes include bidding strategies, firm profits, and platform revenue under symmetric and asymmetric bidding equilibria.

A.2 Derivations and Proofs

A.2.1 Reservation Value

Lemma 1. The reservation value, which determines whether a consumer buys from firm 1 without visiting firm 2 or decides to search further, is obtained by solving for the marginal consumer indifferent between searching and staying.

$$\underbrace{\mathbb{E}[v_2 | v_1 - \Delta p \leq v_2 \leq \bar{v}]}_{\text{Expected benefit from visiting firm 2}} - \underbrace{(v_1 - \Delta p) \cdot [F(\bar{v}) - F(v_1 - \Delta p)]}_{\text{Opportunity cost of visiting firm 2}} \geq \underbrace{s}_{\text{Search cost}}$$

$$\begin{aligned}
\int_{\hat{v}-\Delta p}^{\bar{v}} [(v_2 - p_2) - (\hat{v} - p_1)] f(v_2) dv_2 &= s \\
\int_{\hat{v}-\Delta p}^{\bar{v}} v_2 f(v_2) dv_2 - (\hat{v} - \Delta p) [F(\bar{v}) - F(\hat{v} - \Delta p)] &= s \\
\mathbb{E}[v_2 | \hat{v} - \Delta p \leq v_2 \leq \bar{v}] - (\hat{v} - \Delta p) [F(\bar{v}) - F(\hat{v} - \Delta p)] &= s
\end{aligned}$$

- $\frac{\partial \hat{v}}{\partial s} < 0$. Using Leibniz's rule,^{A.1}

$$\begin{aligned}
0 - (\hat{v} - \Delta p) f(\hat{v} - \Delta p) \frac{\partial \hat{v}}{\partial s} + 0 - \frac{\partial \hat{v}}{\partial s} [1 - F(\hat{v} - \Delta p)] \\
- (\hat{v} - \Delta p) \left[-f(\hat{v} - \Delta p) \frac{\partial \hat{v}}{\partial s} \right] &= 1 \\
\frac{\partial \hat{v}}{\partial s} [-(\hat{v} - \Delta p) f(\hat{v} - \Delta p) - [1 - F(\hat{v} - \Delta p)] + (\hat{v} - \Delta p) f(\hat{v} - \Delta p)] &= 1
\end{aligned}$$

$$\frac{\partial \hat{v}}{\partial s} = -\frac{1}{1 - F(\hat{v} - \Delta p)} < 0$$

- $\frac{\partial \hat{v}}{\partial p_1} > 0$. Using Leibniz's rule,

$$\begin{aligned}
0 - (\hat{v} - \Delta p) f(\hat{v} - \Delta p) \left(\frac{\partial \hat{v}}{\partial p_1} - 1 \right) + 0 - \left(\frac{\partial \hat{v}}{\partial p_1} - 1 \right) [1 - F(\hat{v} - \Delta p)] \\
- (\hat{v} - \Delta p) \left[-f(\hat{v} - \Delta p) \left(\frac{\partial \hat{v}}{\partial p_1} - 1 \right) \right] &= 0 \\
\left(\frac{\partial \hat{v}}{\partial p_1} - 1 \right) [-(\hat{v} - \Delta p) f(\hat{v} - \Delta p) - [1 - F(\hat{v} - \Delta p)] + (\hat{v} - \Delta p) f(\hat{v} - \Delta p)] &= 0
\end{aligned}$$

$$\frac{\partial \hat{v}}{\partial p_1} = 1 > 0$$

^{A.1}When both the function $f(x, t)$ and its partial derivative $f_x(x, t)$ are continuous in both x and t ,

$$\frac{d}{dx} \left(\int_{a(x)}^{b(x)} f(x, t) dt \right) = f(x, b(x)) \cdot \frac{d}{dx} b(x) - f(x, a(x)) \cdot \frac{d}{dx} a(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt$$

- $\frac{\partial \hat{v}}{\partial p_2} < 0$. Using Leibniz's rule,

$$\begin{aligned}
0 - (\hat{v} - \Delta p)f(\hat{v} - \Delta p)\left(\frac{\partial \hat{v}}{\partial p_2} + 1\right) + 0 - \left(\frac{\partial \hat{v}}{\partial p_2} + 1\right)[1 - F(\hat{v} - \Delta p)] \\
- (\hat{v} - \Delta p) \left[-f(\hat{v} - \Delta p)\left(\frac{\partial \hat{v}}{\partial p_2} + 1\right) \right] = 0 \\
\left(\frac{\partial \hat{v}}{\partial p_1} + 1\right) [-(\hat{v} - \Delta p)f(\hat{v} - \Delta p) - [1 - F(\hat{v} - \Delta p)] + (\hat{v} - \Delta p)f(\hat{v} - \Delta p)] = 0
\end{aligned}$$

$$\frac{\partial \hat{v}}{\partial p_2} = -1 < 0$$

■

Given $v_k \sim U[0, V]$.

$$\begin{aligned}
l(\hat{v}, s, \mathbf{p}) = 0 &= \int_{\hat{v}-\Delta p}^V (v_2 - \hat{v} + \Delta p)f(v_2)dv_2 - s \quad [f(v_2) > 0 \implies 0 < \hat{v} - \Delta p < V] \\
&\implies \frac{1}{V} \left[\frac{v_2^2}{2} - \hat{v}v_2 + \Delta p v_2 \right]_{\hat{v}-\Delta p}^V - s = 0 \\
&\implies \frac{1}{V} \left(\frac{V^2}{2} - \hat{v}V + \Delta p V + \frac{\hat{v}^2}{2} - \Delta p \hat{v} + \frac{\Delta p^2}{2} \right) - s = 0 \\
&\implies \frac{\hat{v}^2}{2} - \hat{v}(V + \Delta p) + \frac{V^2}{2} + \Delta p V + \frac{\Delta p^2}{2} - Vs = 0
\end{aligned}$$

Quadratic roots:

$$\begin{aligned}
\hat{v} &= V + \Delta p \pm \sqrt{(V + \Delta p)^2 - 2 \left(\underbrace{\frac{V^2}{2} + \Delta p V + \frac{\Delta p^2}{2}}_{\left(\frac{V+\Delta p}{\sqrt{2}}\right)^2} - Vs \right)} \\
&\implies \hat{v} = V + \Delta p \pm \sqrt{2Vs}
\end{aligned}$$

Deriving constraints for $0 < \hat{v} < V$, we get

$$\hat{v} = \begin{cases} V + \Delta p - \sqrt{2Vs} & \text{when } \Delta p \in [\sqrt{2Vs} - V, \sqrt{2Vs}] \\ V + \Delta p + \sqrt{2Vs} & \text{when } \Delta p \in [-\sqrt{2Vs} - V, -\sqrt{2Vs}] \end{cases}$$

Since the two constraints on Δp do not overlap for non-zero values of s, V , we have a unique value for $\hat{v}$ when either of the constraints are satisfied.

But applying the constraint $0 < \hat{v} - \Delta p < V$, we only have

$$\hat{v} = V + \Delta p - \sqrt{2Vs} \text{ when } \Delta p \in [\sqrt{2Vs} - V, \sqrt{2Vs}]$$

A.2.2 Search Incentives, Demand Asymmetry, and Comparative Statics

Some consumers realize a valuation at firm 1 such that $v_1 - p_1 < 0$, preferring the outside option to product 1. If they visit firm 2, they compare its product to zero. Their search incentive depends on whether:

$$\int_{p_2}^{\bar{v}} (v_2 - p_2) f(v_2) dv_2 \geq s.$$

The “Attraction” condition formalizes this as:

$$A := \mathbb{1}\{\mathbb{E}[v_2 | p_2 \leq v_2 \leq \bar{v}] - p_2 \cdot [1 - F(p_2)] \geq s\}. \quad (\text{A.1})$$

Intuitively, when this constraint binds, it imposes an upper bound on the price firm 2 can charge if it wishes to attract consumers who discovered a low valuation at firm 1. For $F = U[0, 1]$, this condition becomes $p_2 < 1 - \sqrt{2s}$. For a general F , the threshold can be expressed as:

$$\underline{s} = \{t \mid \mathbb{E}[v_2 | p_2 \leq v_2 \leq \bar{v}] - p_2 \cdot [1 - F(p_2)] > s \ \forall s < t\}.$$

Search Rule. Combining the above steps, search behavior can be characterized as follows: For consumers who find product 1 affordable ($v_1 > p_1$), they continue searching only if $v_1 < \hat{v}$. For consumers who find product 1 unaffordable ($v_1 < p_1$), their decision is determined by the “Attraction” condition in Equation (A.1). When both conditions fail, consumers exit the market. This rule is similar to the classic result in [Wolinsky \(1986\)](#).

A.2.3 Consumer Demand

Following terminology commonly used in the literature on prominence, D_{11} and D_{22} can be interpreted as *fresh demand*, representing consumers visiting a firm for the first time, while D_{12} corresponds to *returning demand*, as these consumers revisit firm 1 after inspecting firm 2. These components are shaped by different forces, creating an asymmetry in firms' demand. This asymmetry is central to the equilibrium outcome in Theorem 1, and Lemma A.1 provides comparative statics with respect to search cost.

Lemma A.1. *In the region where prices are not constrained by the “Attraction” condition (i.e., when search costs are low), fresh demand for firm 1 increases with search cost $\left(\frac{\partial D_{11}}{\partial s} > 0\right)$, returning demand for firm 1 decreases with search cost $\left(\frac{\partial D_{12}}{\partial s} < 0\right)$, and demand for firm 2 decreases with search cost $\left(\frac{\partial D_{22}}{\partial s} < 0\right)$.*

Lemma A.1. For D_{11} ,

$$\frac{\partial D_{11}}{\partial s} = -f(\hat{v})\frac{\partial \hat{v}}{\partial s} = \frac{f(\hat{v})}{1 - F(\hat{v} - \Delta p)} > 0$$

For D_{12} , using Leibniz's rule,

$$\begin{aligned}\frac{\partial D_{12}}{\partial s} &= F(\hat{v} - \Delta p)f(\hat{v})\frac{\partial \hat{v}}{\partial s} - 0 + 0 \\ &= -\frac{F(\hat{v} - \Delta p)f(\hat{v})}{1 - F(\hat{v} - \Delta p)} < 0\end{aligned}$$

For D_{22} , using Leibniz's rule,

$$\begin{aligned}\frac{\partial D_{22}}{\partial s} &= f(\hat{v})\frac{\partial \hat{v}}{\partial s} - \left[F(\hat{v} - \Delta p)f(\hat{v})\frac{\partial \hat{v}}{\partial s} - 0 + 0\right] \\ &= \frac{-f(\hat{v}) + F(\hat{v} - \Delta p)f(\hat{v})}{1 - F(\hat{v} - \Delta p)} < 0\end{aligned}$$

■

An increase in search cost lowers the reservation value, reducing the likelihood that consumers continue searching. Consequently, more consumers purchase from firm 1 without visiting firm 2, which raises D_{11} . At the same time, fewer consumers visit firm 2, exerting downward pressure on both D_{12} and D_{22} . Among those who still choose to search, valuations at firm 1 are likely to be low. This negatively affects firm 1's returning demand but partially offsets the decline in firm 2's demand. For D_{22} , these opposing forces – fewer visitors overall versus higher conversion among those who search – work in opposite directions. In the region where firm 2 can freely adjust its price (i.e., when Equation (A.1) does not bind), the first effect dominates, leading to

an overall decline in D_{22} .

A.2.4 Equilibrium - No Auction

Random Search: Solving for symmetric eq.

$$\max_p pD(p) = p \frac{D_{11} + D_{12} + D_{22}}{2}$$

For some deviation p and equilibrium price p^{Ran}

$$\begin{aligned} \frac{\partial \pi^{Ran}(p^{Ran})}{\partial p} &= \frac{\partial}{\partial p} (p(1 - pp^{Ran})) \\ &= 1 - 2(p^{Ran})^2 = 0 \\ \implies p^{Ran} &= \frac{1}{\sqrt{2}} \end{aligned}$$

It requires $pf'(p) < 2f(p)$ (or hazard rate is increasing in p) for p^{Ran} to be a global maximum.

Lemma 2. Solving for firm 1 when prices are symmetric,

$$p^* = \sqrt{2 + 2s} - 1$$

Solving for firm 2 when prices are symmetric,

$$p^* = \sqrt{2 - 2s} - 1$$

They are equal only if $s = 0$. ■

Theorem 1. Solving First order conditions, we get

$$\begin{aligned} p_{1,NA}^* &= \frac{1 - F(\widehat{v}^*) + \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f(v_1) \cdot dv_1}{f(\widehat{v}^*) - \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1} \\ p_{2,NA}^* &= \frac{A \cdot [1 - F(p_{2,NA}^*)]F(p_{1,NA}^*) + F(\widehat{v}^*) - F(p_{1,NA}^*) - \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f(v_1) \cdot dv_1}{A \cdot f(p_{2,NA}^*)F(p_{1,NA}^*) + f(\widehat{v}^*) - F(p_{2,NA}^*)f(p_{1,NA}^*) - \int_{p_{1,NA}^*}^{\widehat{v}^*} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1} \end{aligned}$$

For $p_{1,NA}^*, p_{2,NA}^*$ to be global maxima, we require

$$\begin{aligned} p_1 \cdot \left(f'(\widehat{v}) - \int_{p_1}^{\widehat{v}} f(v_1 - \Delta p) \cdot f''(v_1) \cdot dv_1 \right) &\leq 2 \left(f(\widehat{v}) - \int_{p_1}^{\widehat{v}} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right) \\ p_2 \cdot \left(A \cdot f'(p_{2,NA}^*) F(p_{1,NA}^*) - f'(\widehat{v}) - f(p_2) f(p_1) + F(\widehat{v} - \Delta p) \cdot f'(\widehat{v}) - \int_{p_1}^{\widehat{v}} f(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right) \\ &\leq 2 \left(A \cdot f(p_{2,NA}^*) F(p_{1,NA}^*) + f(\widehat{v}) - F(p_2) f(p_1) - \int_{p_1}^{\widehat{v}} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right) \end{aligned}$$

For $p_{1,NA}^*$ and $p_{2,NA}^*$ to be global maxima, the second-order conditions require the profit function to be concave in prices. These conditions hold under standard assumptions such as log-concavity of F , and are trivially satisfied for the Uniform distribution used in the working example of $F = U[0, 1]$. ■

Proposition 1. To find demand:

$$D_{11} = 1 - \frac{\widehat{v}}{V}$$

D_{12} requires $p_1 < v_1 < \widehat{v}$, $v_2 - p_2 < v_1 - p_1$.

$$\begin{aligned} D_{12} &= \int_{p_1}^{\widehat{v}} F_2(v_1 - \Delta p) f(v_1) dv_1 \\ &= \frac{(\widehat{v} - \Delta p)^2}{2V^2} - \frac{p_2^2}{2V^2} \end{aligned}$$

D_{22} requires $v_1 < \widehat{v}$, $v_2 - p_2 > \max\{0, v_1 - p_1\}$.

$$\begin{aligned} D_{22} &= \underbrace{\left(1 - \frac{p_2}{V}\right) \frac{p_1}{V} \cdot \mathbb{1}_{p_2 < \frac{V}{2} - s}}_{v_1 < p_1} + \underbrace{\int_{p_1}^{\widehat{v}} [1 - F_2(v_1 - \Delta p)] f(v_1) dv}_{v_1 > p_1} \\ &= \left(1 - \frac{p_2}{V}\right) \frac{p_1}{V} + \frac{(\widehat{v} - p_1)}{V} - \frac{(\widehat{v} - \Delta p)^2}{2V^2} + \frac{p_2^2}{V^2} \end{aligned}$$

Profit Maximisation:

$$\begin{aligned} p_1 &= \frac{V}{4} + \frac{s}{2} + \frac{p_2}{2} - \frac{p_2^2}{4V} \\ p_2 &= \frac{1}{3} \left(2p_1 + 2V - \sqrt{10p_1^2 + 2p_1V + 6sV + V^2} \right) \end{aligned}$$

- At $\underline{s}_{NA}$, $\hat{v} = p_{1,NA}^*$. Solving for s using the equations for prices, we get $\underline{s}_{NA} = \frac{1}{4}$.

•

$$\frac{dp_{1,NA}^*}{ds} = \frac{1 + (1 - p_{2,NA}^*) \frac{dp_{2,NA}^*}{ds}}{2}$$

$$\frac{dp_{2,NA}^*}{ds} = \frac{(1 - p_{2,NA}^*) \left(-1 + \frac{dp_{1,NA}^*}{ds} \right)}{\underbrace{2(1 + \Delta p_{NA}^*)^2 - D_{22}^*}_{>0}}$$

Note that $(1 - p_2) \geq 1 - p^{Ran}$ and $D_{22} \leq \frac{1 - (p^{Ran})^2}{2}$ whose bounds are realised when there is random search. This implies that the denominator of the RHS of the above expression is bounded below by zero.

Substituting this result in the previous expression, we get

$$\frac{dp_{1,NA}^*}{ds} = \frac{1 + (1 - p_{2,NA}^*) \frac{(1 - p_{2,NA}^*) \left(-1 + \frac{dp_{1,NA}^*}{ds} \right)}{2(1 + \Delta p_{NA}^*)^2 - D_{22}^*}}{2}$$

Note that

$$\begin{aligned} & \frac{(1 - p_{2,NA}^*)^2}{2(1 - p_{2,NA}^*)^2 - D_{22}^*} \\ &= \frac{1}{2 - \underbrace{\frac{D_{22}^*}{(1 - p_{2,NA}^*)^2}}_{\in [0,1]}} < 1 \end{aligned}$$

$$\Rightarrow \frac{dp_{1,NA}^*}{ds} \in (0, 1) , \quad \frac{dp_{2,NA}^*}{ds} < 0$$

- Since prices are continuous and p_1 is monotonically increasing while p_2 is monotonically decreasing, they cross each other at one point atmost (which happens at $s = 0$).
- At the lower bound, $s = 0$, ordered consumer search is equivalent to that of random search

as shown below. At $s = 0$,

$$p_{1,NA}^* = p_{2,NA}^* = \frac{1 - (p_{NA}^*)^2}{2} = p^{Ran} = \sqrt{2} - 1$$

At $p_{2,NA}^* = 0$, $F(\hat{v}^*) - F(p_{1,NA}^*) - \int_{p_1}^{\hat{v}^*} F(v_1 - \Delta p) \cdot f(v_1) \cdot dv_1 = 0$. This implies that

$$p_{1,NA}^* = 1 - p_{1,NA}^* = p^{Mon} = \frac{1}{2}$$

This upper bound for $p_{1,NA}^*$ is hit at $(\underline{s})_{NA} = \frac{1}{4}$. For $s > \underline{s}$, condition A binds

$$p_2 < 1 - \sqrt{2}s$$

Therefore, $\bar{s} = \frac{1}{2}$.

■

This following exercise provides the formal welfare analysis for the benchmark case without auction, using the working example of $F = U[0, 1]$. Figure A.2 illustrates Consumer Surplus (CS_{NA}), Industry Revenue (IR_{NA}), and Total Welfare (TW_{NA}) as functions of search cost.

As search cost increases, searches become costlier and overall consumer surplus declines. Consumers who buy from firm 1 without searching (D_{11}) avoid this cost, and their share rises with s , contributing positively to CS_{NA} . Conversely, consumers who search and return to firm 1 (D_{12}) incur the search cost, but their importance diminishes as s increases. Firm 2's price falls with search cost due to posted prices, partially offsetting the negative effect of search cost on its visitors.

In region II, market segmentation occurs: firm 1's situation remains constant, while firm 2 faces a constraint that forces it to lower its price as s rises. This benefits consumers until firm 2 exits the market in region III. Overall, TW_{NA} peaks at a lower search cost than IR_{NA} , reflecting a trade-off between efficiency and industry revenue.

Efficiency Measures: Conversion rates provide a measure of positional performance:

$$(q_1)_{NA} = D_{11} + D_{12},$$

$$(q_2)_{NA} = \frac{D_{22}}{1 - D_{11}}.$$

As s increases, firm 1's conversion rate rises because consumers prefer the option in hand, while firm 2's conversion rate also increases among those who search, culminating in full conversion as $p_2 \rightarrow 0$ at $s = \bar{s}$.

Total Transactions: A measure of overall market efficiency is the total volume of transactions:

$$TR = D_{11} + D_{12} + D_{22} = 1 - p_1 p_2.$$

Total transactions reach a maximum at $s = \bar{s}$, driven by sales at firm 2. These efficiency measures provide clear empirical predictions for testing the model.

A.2.5 Equilibrium With Auction

Symmetric bids

Symmetric Bidding. Let η_k denote the total commission paid by firm k . Assume there exists a symmetric equilibrium where $b_i = b_j = b_{sym}$. In this case, each firm gets the first position with probability. Profit for each firm is

$$\pi_i = \pi_j =: \pi = \frac{Rev_1 + Rev_2}{2} - \frac{\eta_1 + \eta_2}{2}$$

where

$$\begin{aligned}\eta_1 &= b_{sym} \\ \eta_2 &= \widehat{b}(1 - D_{11})\end{aligned}$$

Individual Rationality: Firms need to make non-negative profits.

$$\begin{aligned}\pi &\geq 0 \\ \iff b_{sym} &\leq Rev_1 + Rev_2 - \widehat{b}(1 - D_{11})\end{aligned} \tag{A.2}$$

Therefore, $b_{sym} \in [\widehat{b}, Rev_1 + Rev_2 - \widehat{b}(1 - D_{11})]$. For any value in this set, consider a deviation.

Case One: Let $b_i > b_j = b_{sym}$ be the bids of the two firms. Assume that the positions assigned are $i \rightarrow 1, j \rightarrow 2$.

Individual Rationality: Firm needs to make non-negative profit.

$$\begin{aligned}\pi_1 &\geq 0 \\ \iff Rev_1 &\geq b_j\end{aligned}$$

Incentive Compatibility: Firm deviates only if there is a gain from doing so.

$$\begin{aligned}
& \pi_1 \geq \pi = \frac{\pi_1 + \pi_2}{2} \\
\iff & \frac{\pi_1}{2} \geq \frac{\pi_2}{2} \\
\iff & \pi_1 \geq \pi_2 \\
\iff & Rev_1 - b_j \geq Rev_2 - \widehat{b}(1 - D_{11}) \\
\iff & b_j \leq Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11}))
\end{aligned} \tag{A.3}$$

Combining (A.2) and (A.3), there exists a deviation if

$$\begin{aligned}
& Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})) > Rev_1 + (Rev_2 - \widehat{b}(1 - D_{11})) \\
\iff & Rev_2 - \widehat{b}(1 - D_{11}) < 0 \\
\iff & \widehat{b} > \frac{Rev_2}{1 - D_{11}}
\end{aligned}$$

This gives the sufficient condition to find a deviation. Then, we see that there exists no deviation of the kind $b_i > b_j$ iff

$$b_j \in \left[Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})), Rev_1 + (Rev_2 - \widehat{b}(1 - D_{11})) \right] \quad \text{and} \quad \widehat{b} \leq \frac{Rev_2}{1 - D_{11}} \tag{A.4}$$

Case Two: Consider the deviation $b_i < b_j$ and position assignments $i \rightarrow 2, j \rightarrow 1$. Following similar steps, we find that there exists no such deviation iff

$$b_j \in \left[0, Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})) \right] \quad \text{and} \quad \widehat{b} \leq \frac{Rev_2}{1 - D_{11}} \tag{A.5}$$

Combining (A.4), and (A.5), we get the optimal symmetric bid

$$b_i = b_j = Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})) \tag{A.6}$$

This is feasible only when

$$\widehat{b} \leq \frac{Rev_2}{1 - D_{11}} \tag{A.7}$$

Asymmetric bids

Consider the bids $b_i > b_j$ and position assignments $i \rightarrow 1, j \rightarrow 2$.

Individual Rationality: Firms need to make non-negative profits.

$$\begin{aligned} \pi_1 &\geq 0 \\ \iff b_j &\leq Rev_1 \end{aligned} \tag{A.8}$$

$$\begin{aligned} \pi_2 &\geq 0 \\ \iff \widehat{b} &\leq \frac{Rev_2}{1 - D_{11}} \end{aligned} \tag{A.9}$$

Incentive Compatibility: Firm i deviates to position 2 iff there is a gain from doing so.

$$\begin{aligned} Rev_1 - \eta_1 &\geq Rev_2 - \eta_2 \\ \iff Rev_1 - b_j &\geq Rev_2 - \widehat{b}(1 - D_{11}) \\ \iff b_j &\leq Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})) \end{aligned} \tag{A.10}$$

Firm j doesn't want position 1, given the above bids.

$$\begin{aligned} Rev_1 - \widetilde{\eta}_1 &\leq Rev_2 - \eta_2 \\ \iff Rev_1 - b_i &\leq Rev_2 - \widehat{b}(1 - D_{11}) \\ \iff b_i &\geq Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})) \end{aligned} \tag{A.11}$$

Combining (A.10) and (A.11), we get the optimal asymmetric bids

$$b_i \in \left(Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})), \infty \right) \tag{A.12}$$

$$b_j \in \left[\widehat{b}, Rev_1 - (Rev_2 - \widehat{b}(1 - D_{11})) \right] \tag{A.13}$$

This is feasible only when

$$\widehat{b} \leq \min \left\{ \frac{Rev_1 - Rev_2}{D_{11}}, \frac{Rev_2}{1 - D_{11}} \right\} \tag{A.14}$$

Auction reserve price

The intermediary's objective is to maximise its revenue.

$$\max_{\widehat{b}} Rev_m = b_j + \widehat{b}(1 - D_{11})$$

Note that

$$\frac{\partial \left[Rev_1 - \left(Rev_2 - \widehat{b}(1 - D_{11}) \right) \right]}{\partial \widehat{b}} \geq 0$$

This implies that b_j would take larger values when $\widehat{b}$ increases for during both symmetric and asymmetric bidding.

$$\frac{\partial Rev_m}{\partial \widehat{b}} \geq 0$$

Theorem 2. Prices are obtained by solving the first-order conditions for equation 6. For p_1^*, p_2^* to be global maxima:

$$\begin{aligned} p_1 \cdot \left(f'(\widehat{v}) - \int_{p_1}^{\widehat{v}} f(v_1 - \Delta p) \cdot f''(v_1) \cdot dv_1 \right) &\leq 2 \left(f(\widehat{v}) - \int_{p_1}^{\widehat{v}} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right) \\ p_2 \cdot \left(A \cdot f'(p_{2,NA}^*) F(p_{1,NA}^*) - f'(\widehat{v}) - f(p_2) f(p_1) + F(\widehat{v} - \Delta p) \cdot f'(\widehat{v}) - \int_{p_1}^{\widehat{v}} f(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right) \\ &\leq 2 \left(A \cdot f(p_{2,NA}^*) F(p_{1,NA}^*) f(\widehat{v}) - F(p_2) f(p_1) - \int_{p_1}^{\widehat{v}} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right) - r_2 f'(\widehat{v}) \end{aligned}$$

For p_1^* and p_2^* to be global maxima, the second-order conditions require the profit function to be concave in prices. These conditions hold under standard assumptions such as log-concavity of F , and are trivially satisfied for the Uniform distribution used in the working example of $F = U[0, 1]$. ■

Proposition 2.

$$\begin{aligned} \frac{dp_1^*}{ds} &= \frac{1 + (1 - p_2^*) \frac{dp_2^*}{ds}}{2} \\ \frac{dp_2^*}{ds} &= \frac{(1 - p_2^*) \left(-1 + \frac{dp_1^*}{ds} + \frac{d\widehat{b}^*}{ds} \right)}{\underbrace{2(1 + \Delta p^*)^2 - D_{22}^*}_{>0}} \end{aligned}$$

Note that $(1 - p_2) \geq 1 - p^{Ran}$ and $D_{22} \leq \frac{1 - (p^{Ran})^2}{2}$ whose bounds are realised when there is random search. This implies that the denominator of the RHS of the above expression is bounded below by zero.

Substituting this result in the previous expression, we get

$$\frac{dp_1^*}{ds} = \frac{1 + (1 - p_2^*) \frac{(1 - p_2^*) \left(-1 + \frac{dp_1^*}{ds} + \frac{d\widehat{b}^*}{ds} \right)}{2(1 + \Delta p^*)^2 - D_{22}^*}}{2}$$

Note that

$$\begin{aligned} & \frac{(1 - p_2^*)^2}{2(1 - p_2^*)^2 - D_{22}^*} \\ &= \frac{1}{\underbrace{2 - \frac{D_{22}^*}{(1 - p_2^*)^2}}_{\in [0,1)}} \in \left[\frac{1}{2}, 1 \right) \\ \Rightarrow \frac{dp_1^*}{ds} &= \frac{1 + \left(-1 + \frac{d\widehat{b}^*}{ds} \right) \frac{(1 - p_2^*)^2}{2(1 - p_2^*)^2 - D_{22}^*}}{2 \times \left[\frac{3}{2}, 2 \right)} \\ \Rightarrow \frac{dp_1^*}{ds} &\in \left(0, \frac{1}{3} \frac{d\widehat{b}^*}{ds} \right) \end{aligned}$$

$$\frac{dp_2^*}{ds} = \frac{(1 - p_2^*) \left(-1 + \frac{dp_1^*}{ds} + \frac{d\widehat{b}^*}{ds} \right)}{2(1 + \Delta p^*)^2 - D_{22}^*} < 0 \quad \text{for } \frac{d\widehat{b}^*}{ds} \leq \frac{3}{2}$$

Assuming the above condition to be true,

$$\begin{aligned} \frac{dRev_1^*}{ds} &= p_1^* \left(\frac{dp_1^*}{ds} \right) > 0 \\ \frac{dRev_2^*}{ds} &= p_2^* \left(\underbrace{-1 - (1 - p_2^*) \frac{dp_2^*}{ds}}_{=-2 \frac{dp_1^*}{ds}} \right) < 0 \\ \Rightarrow \frac{d\widehat{b}^*}{ds} &= \begin{cases} \frac{dRev_1^* - Rev_2^*}{ds} \in [0, \frac{3}{4}] & \text{if } s < \widehat{s} \\ \frac{dRev_2^*}{ds} \text{ to do} & \text{if } s > \widehat{s} \end{cases} \end{aligned}$$

This result for $s < \hat{s}$ fulfils the sufficient condition assumed above. ■

Corollary 2.

$$\frac{dp_1^*}{ds} > \frac{dp_{1,NA}^*}{ds} \quad , \quad \frac{dp_2^*}{ds} > \frac{dp_{2,NA}^*}{ds}$$

■

B Additional Discussion of Model

B.1 Are the equilibria self-fulfilling?

Following the early work in [Stahl \(1989\)](#), treating the first search as free has been a standard assumption in the consumer search literature.^{B.1} Moreover, since consumers encounter online ads free of charge, their search costs are mainly the amount of time spent browsing the product on the Internet. In this context, the assumption of free visit to the prominent firm is motivated by ad listings where websites often host a larger advertisement of a particular firm with product details while the others are shown as smaller thumbnails. Therefore, options may be presented to consumers in an exogenously restricted order, and it costs lesser for consumers to learn the details of the first firm. Another motivating example is that of websites that make some ads less prominent by placing them at a less immediate location, requiring the consumer to scroll down or click on a hyperlink, thus entailing a search cost to visit the second firm. To extend the fit of my model to more general settings, I consider alternative assumptions in this subsubsection using the working example of $F = U[0, 1]$.

If $s > 0$ for the prominent as well? Note that consumers still have to follow the displayed order. A positive search cost for firm 1 imposes an additional constraint which determines what fraction of the population prefers to visit firm 1 rather than choose the outside option of zero. Formally, it can be represented as

$$\int_{p_1}^1 (v_1 - p_1) f(v_1) dv_1 - s > 0$$

This constraint does not bind when $p_1 < 1 - \sqrt{2s}$. For low search cost (region I), the equilibrium prices are the same as in figure 3 as the above constraint does not bind. See Online Appendix Section D for further discussion on costly first search.

^{B.1}Recent work on consumer search has often assumed first search as free. See, for instance, [Janssen, Moraga-González, and Wildenbeest \(2007\)](#), [Janssen and Parakhonyak \(2013\)](#), [Ding and Zhang \(2018\)](#), [Janssen and Shelegia \(2020\)](#).

If given a choice, do consumers search in the displayed order? For low search cost (region I), p_2 is greater than p_1 and consumers prefer to follow the order in which firms are displayed.

At $s = \underline{s}$, firms charge an identical price, $p_1 = p_2$. Hence, consumers are indifferent between starting from either firm and if consumers can choose which firm to visit, we will have random search. Thus, firms' prices will follow Wolinsky (1986) and have a value of $1 - \sqrt{2}s$ for $\underline{s} < s < \bar{s}$. At $s = \underline{s} = \frac{1}{8}$, this value coincides with the monopoly price, $\frac{1}{2}$.

B.2 Superstar and Fringe

In my model, the intermediary would prefer to design the market at a higher search cost than the welfare-maximising level. Also, note that firms' revenues get more dispersed with search cost. Another factor that affects revenue dispersion is the per-click commission cost. This asymmetry in cost structure significantly changes firms' price setting problem, as captured starkly in the duopoly model. Potentially, a superstar firm may have sufficient funds to *enter* the top positions by paying the *fixed cost*. This pushes the fringe firms to lower positions. Since the profits are distributed in order of position, this result highlights a mechanism through which a profit-maximising intermediary may exacerbate the gap between superstar and fringe firms and provide a rationale for growing market concentration. See Online Appendix Section E, F, and H for a discussion the timing of the game and auction format respectively.

C Extensions

C.1 Platform Revenue Models and Welfare

The baseline model analyzed an intermediary revenue structure based on cost per click (CPC), implemented via a Generalized Second Price (GSP) auction. Firms bid for ad positions and pay CPC for each click received. This design reduces consumer surplus due to the asymmetric pass-through of CPC costs into product prices. I now consider alternative revenue models that maintain ranking but may relax this asymmetry.

First, I examine two alternatives: (i) a cost per purchase (CPP) model, where the intermediary charges a fee per purchase, and (ii) a *consumer subscription fee* model, where consumers pay a fixed fee ex-ante.^{C.1} Both models alter incentives and have distinct welfare implications. Second, I analyze a combined model where the intermediary earns revenue from both CPC and consumer subscription fees, introducing a trade-off between reducing CPC charges and extracting surplus from consumers. Finally, for completeness, I study a Generalized First Price auction in Online

^{C.1}A random display-structure of ads is a less interesting model to study for two reasons: (i) it eliminates one of the main benefits of the intermediary, which is to use the data collected to recommend more suitable ads, and (ii) while an intermediary may claim to randomize its listings, it may continue to do some selection and this is hard to monitor for firms and policymakers.

Appendix Section H, leaving the CPC design unchanged. Outcomes coincide with those under GSP.

C.1.1 Cost Per Purchase

I now examine the implications of replacing the platform's CPC model with a CPP model, where the intermediary charges a fee per purchase rather than per click. Under CPC, the platform's incentive is tied to generating clicks, whereas CPP aligns revenue with completed transactions. This change significantly alters bidding incentives and competitive outcomes.

In equilibrium, the intermediary sets the auction reserve price $\hat{b}$ equal to the anticipated price p_2 , thereby extracting all surplus from the non-prominent firm. This effectively raises firm 2's marginal cost to the reservation value $\hat{v}$, limiting its ability to compete. As a result, firm 2 can only attract a positive measure of consumers if the intermediary prefers a duopoly over monopoly (à la tie-breaking). And when $p_2 = \hat{v}$, the market is segmented for any $s > 0$, that is consumers stop comparing product prices. Consequently, the prominent firm charges the monopoly price, amplifying its competitive advantage relative to the CPC model.^{C.2} This result is formally presented in Proposition C.1 and Corollary C.1.

Proposition C.1. *In equilibrium, $\hat{b} = p_2 = \hat{v}$. Search occurs only among consumers who cannot afford product 1. Consequently, the market is segmented for any $s > 0$ (i.e., $\underline{s} = 0$), and firm 1 charges the monopoly price.*

Corollary C.1. *The cost-per-purchase (cpp) commission structure makes the firm outcomes more unequal than the cost-per-click (cpc) commission structure. Thus, the prominent firm focuses its sales on fewer, but more valuable, consumers and extracts more rent from them.*

$$\begin{aligned} p_1^{cpp} &\geq p_1^{cpc} \\ p_2^{cpp} &\geq p_2^{cpc} \\ Rev_1^{cpp} &\geq Rev_1^{cpc} \\ Rev_2^{cpp} &\leq Rev_2^{cpc} \end{aligned}$$

Welfare implications: CPP incentivizes the intermediary to prioritize purchases rather than clicks, reducing consumer-surplus loss from excessive search compared to CPC. However, fewer searchers also reduce total transactions and concentrate market power with the prominent firm, lowering industry revenue and overall competitiveness. Except at very low search costs, the first effect dominates, making consumers better off while widening inequality between firms. Figure

^{C.2}This result parallels Petrikaitė (2018), where a multi-product monopolist sets search costs to induce segmentation. The CPP structure creates a similar effect by raising marginal costs for the non-prominent firm.

C.1 illustrates this trade-off using the working example of $F = U[0, 1]$.

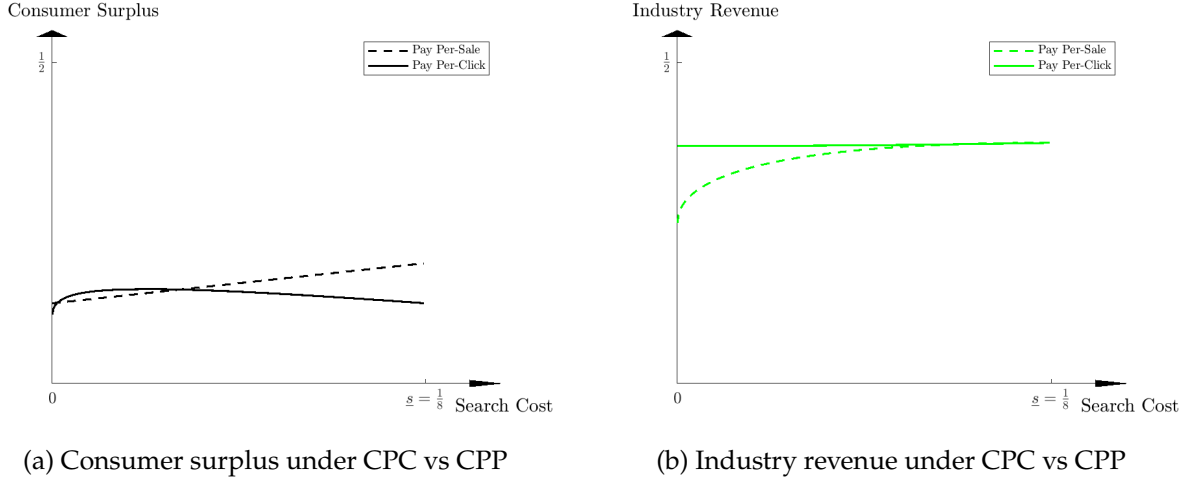


Figure C.1: Welfare trade-offs between Cost Per Click (CPC) and Cost Per Purchase (CPP): CPP improves consumer surplus at low search costs but reduces industry revenue.

C.1.2 Consumer Subscription

I now examine the welfare implications of replacing the platform's CPC model with a *consumer subscription fee* model. Under this design, consumers pay a fixed fee ex-ante, prior to making any purchase, and internalize this cost when deciding whether to search and buy. Formally, consumers draw valuations from:

$$v_1, v_2 \sim U[0, 1 - sub],$$

where sub denotes the homogeneous subscription fee paid before purchase. Figure C.2 plots consumer surplus (CS) and industry revenue (IR) for $F = U[0, 1]$ at $\underline{s} = \frac{1}{8}$ across different subscription levels.

As the subscription fee increases, consumers' maximum willingness to pay falls, leading to lower product prices. Combined with fewer searchers (similar to the CPP model), this effect raises consumer surplus despite reducing total transactions. Consequently, industry revenue declines, making this model consumer-surplus improving but at the expense of the intermediary's revenue.

Figure C.3 compares alternative platform revenue models for $F = U[0, 1]$ at $\underline{s} = \frac{1}{8}$. Moving down the panels, consumer surplus increases as a larger share of total surplus shifts from the intermediary to consumers. From a static perspective, this suggests that policymakers might

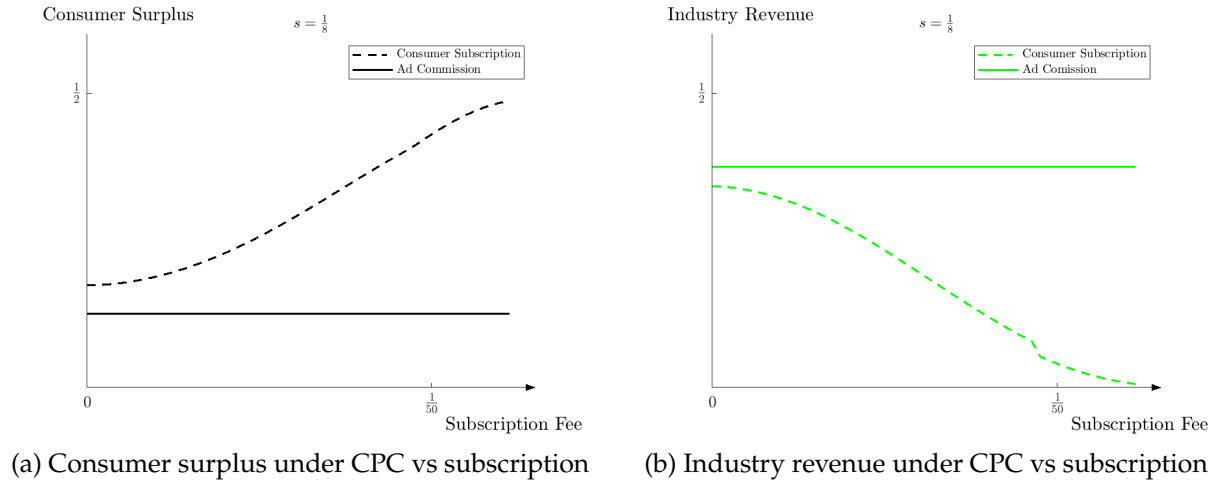


Figure C.2: Welfare trade-offs between Cost Per Click (CPC) and consumer subscription: Subscription increases consumer surplus but reduces industry revenue.

prefer these alternatives. However, the same shift is accompanied by a decline in total transactions and a rise in market concentration (HHI), indicating weaker long-run competition. Thus, the socially optimal mechanism depends on the weight placed on static welfare gains versus dynamic considerations.

C.1.3 Combining Platform Revenue Models

Finally, I consider a hybrid design where the intermediary draws revenue from two sources: (i) advertisement charges determined through a Generalized Second Price (GSP) auction based on CPC, and (ii) a consumer subscription fee paid ex-ante. This structure introduces a trade-off for the platform: reducing CPC charges increases consumer surplus (and thus the potential subscription revenue) but simultaneously lowers advertising income. The intermediary's optimal strategy depends on balancing these two margins—ad revenue versus consumer-side surplus extraction.

The intermediary faces a trade-off: while lowering the subscription fee can increase consumer surplus and thus potential revenue from subscriptions, enabling this new source requires reducing its existing source, that is the advertising revenue. This adjustment operates through two channels. At the extensive margin, consumers who do not purchase generate no surplus to share with the intermediary. At the intensive margin, reducing CPC charges lowers product prices, leaving more surplus with consumers, which the intermediary can partially appropriate through subscriptions.

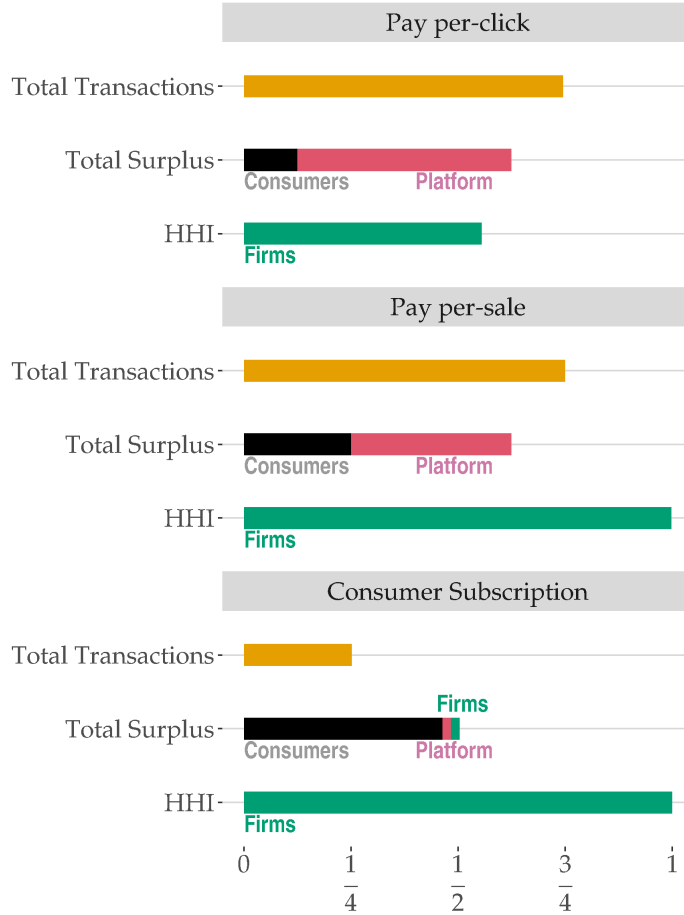


Figure C.3: Comparison of platform revenue models: Consumer surplus, industry revenue, and market concentration

Given this trade-off, the intermediary seeks to increase purchases at the non-prominent position, where consumers typically retain higher surplus due to lower prices under the asymmetric CPC structure (see Section 4). To achieve this, the intermediary reduces the auction reserve price $\hat{b}$, raising profits for the non-prominent firm. This, in turn, weakens incentives for firms to compete for prominence and reduces CPC revenue from the top position, leading to an overall decline in advertising income.

Figure C.4 illustrates the trade-off faced by the intermediary when combining CPC-based ad revenue with consumer subscription fees. For $F = U[0, 1]$, the rate of increase in consumer surplus from lowering CPC charges is smaller than the corresponding decline in advertising revenue (a result robust to different search costs). Consequently, a revenue-maximizing platform prefers to rely solely on CPC income, even when allowed to appropriate a share of consumer

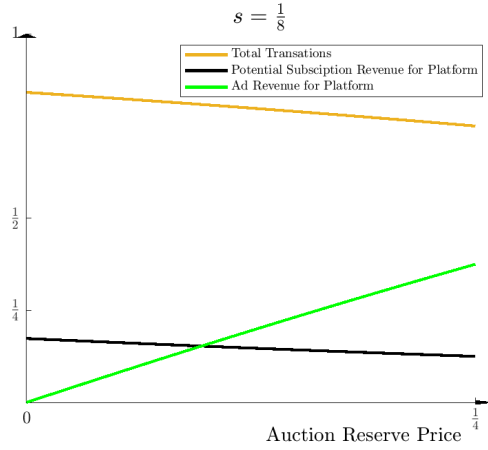


Figure C.4: Platform's revenue trade-off: Ad revenue versus consumer subscription

surplus. This finding provides additional rationale for cross-subsidization strategies observed in two-sided markets.^{C.3} In competitive environments with multiple platforms, the incentive to forgo subscription fees may strengthen, though the net effect remains ambiguous and is beyond the scope of this paper.

C.2 Asymmetric Firms

Firms competing in position auctions may differ in quality or relevance. To capture this, I relax the symmetry assumption and introduce vertical differentiation: firms vary by the maximum valuation a consumer can realize at each firm, which I refer to as *quality*.

Theorem C.2 characterizes the equilibrium under asymmetry. Table C.1 summarizes outcomes for different auction reserve prices $\hat{b}$. Without loss of generality, let firm i have higher quality than firm j , generating greater revenue under identical conditions. The reservation value for the search rule becomes:

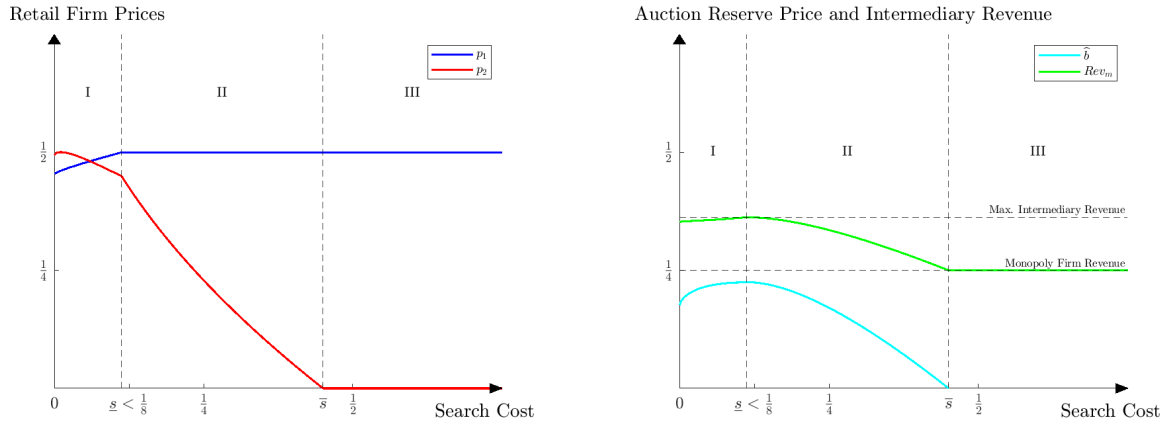
$$\tilde{v}_\alpha = \alpha + \Delta p - \sqrt{2\alpha s}.$$

Proposition C.2. *There exists no symmetric equilibrium. In the unique asymmetric equilibrium in undominated pure strategies, the higher-quality firm i secures the prominent position by bidding more aggressively. Optimal bids and reserve price satisfy:*

$$\hat{b} = \frac{Rev_{2,j}}{1 - D_{11,i}}, \quad b_i \in (Rev_{1,j} - \pi_{2,j}, \infty), \quad b_j = Rev_{1,i} - \pi_{2,i}.$$

Intuition. The absence of symmetric equilibrium reflects the compounding effect of positional

^{C.3}See, for instance, Baye and Morgan (2001), Caillaud and Jullien (2003), and Chen and Rey (2019).



(a) Prices: Higher quality shifts price levels upward for the prominent firm.

(b) Commissions: Higher quality increases CPC and intermediary revenue.

Figure C.5: Equilibrium outcomes with auction under quality asymmetry.

and quality asymmetries. Higher quality amplifies the advantage of prominence: firm i bids higher, earns more, and faces less competitive pressure. Figure C.5 illustrates equilibrium prices and commissions when firms differ in quality, assuming valuations $v_1 \sim U[0, 1]$ and $v_2 \sim U[0, \alpha]$ for $\alpha > 0$.

In Online Appendix G, I extend the analysis to horizontal differentiation, where firms differ by *relevance* – the maximum measure of customers they can attract, for e.g., due to brand loyalty. Results are qualitatively similar: lower relevance or quality reduces firm j 's ability to compete, amplifying its disadvantage. Consequently, the intermediary prefers monopoly at a lower search cost than in the symmetric case ($\bar{s} < 1/2$), as firm j 's revenue potential exhausts earlier.

Table C.1: Equilibrium outcomes for different auction reserve prices under asymmetric firms

Auction Reserve Price	Firms find ...	Firms' Bids	Firms' Profits
$\widehat{b} > Rev_{1,j}$	neither feasible*	$b_1 = \times$ $b_2 = \times$	$\pi_1 = 0$ $\pi_2 = 0$ $Rev_m = 0$
$\max \left\{ \frac{Rev_{1,i} - Rev_{2,i}}{D_{11,j}}, \frac{Rev_{2,j}}{1 - D_{11,i}} \right\} < \widehat{b} < Rev_{1,j}$	position 1 feasible	$b_1 = \frac{Rev^{Mon}}{2}$ $b_2 = \frac{Rev^{Mon}}{2}$	$\pi_1 = 0$ $\pi_2 = 0$ $Rev_m = \frac{Rev^{Mon}}{2}$
$\frac{Rev_{2,j}}{1 - D_{11,i}} < \widehat{b} < \frac{Rev_{1,i} - Rev_{2,i}}{D_{11,j}}$	position 1 feasible	$b_1 = \frac{Rev^{Mon}}{2}$ $b_2 = \frac{Rev^{Mon}}{2}$	$\pi_1 = 0$ $\pi_2 = 0$ $Rev_m = \frac{Rev^{Mon}}{2}$
$\widehat{b} \in \left[0, \min \left\{ \frac{Rev_{1,i} - Rev_{2,i}}{D_{11,j}}, \frac{Rev_{2,j}}{1 - D_{11,i}} \right\} \right]$	both feasible asymmetric bids	$b_1 \in (Rev_{1,j} - \pi_{2,j}, \infty)$ $b_2 \in [\widehat{b}, Rev_{1,i} - \pi_{2,i}]$	$\pi_1 = Rev_{1,i} - b_2$ $\pi_2 = Rev_{2,j} - \widehat{b}(1 - D_{11,i})$ $Rev_m = b_2 + \widehat{b}(1 - D_{11,i})$

Notes: Rev_k denotes revenue of firm in position k . Feasibility refers to non-negative profits. Rev^{Mon} is monopoly revenue. Outcomes include bidding strategies, firm profits, and platform revenue under asymmetric equilibrium.

*Neither feasible indicates zero profits for both firms.

D Appendix: Empirical Analysis

Criteo is one of the largest players^{D.1} in the e-commerce marketing sector. It works with digital platforms and retailers/brands to help serve online advertisements to consumers. Some of its main functions are to design bidding strategies for retailers in order to procure display positions, provide infrastructure to intermediaries in order to conduct real-time auctions, etc. Criteo continuously collects a large amount of data on consumers' online behaviour and firms' advertisement characteristics to understand when searchers convert to buyers and how firms bid.^{D.2} Hence, this user-level data provides a unique opportunity to explore the interaction between consumers and firms in the ad auction market.

D.1 Descriptive Statistics

Table D.1 reports click and purchase behavior across positions. Approximately half of users do not click on any ad, consistent with the notion that display ads target consumers at early stages of the purchase process. Among those who click, the order of clicks strictly follows the order of display, supporting the assumption of ordered search in the theoretical model.

^{D.1}Criteo generated close to \$1B in annual revenue in 2017 in USA, the year in which this data was released. (International Data Corporation (IDC) [report](#) 2017. Accessed May 18, 2021)

^{D.2}Criteo has collected data on more than 2.5B users and 3500 products, by 2021. (<https://criteo.investorroom.com>, accessed May 18, 2021.)

Table D.1: Prominence and user behavior: Click and conversion patterns

Activity	# Users	Users (%)
Click		
See both ads	985,273	100.00
Visit none	508,294	51.59
Visit only first ad	368,186	37.37
Visit second ad only	0	0.00
Visit both ads	108,793	11.04
Conversion		
Visit only first ad, buy from first	12,109	2.61
Visit both ads, buy from first	6,888	1.49
Visit both ads, buy from second	4,921	1.06
Visit both ads, buy none	439,899	94.84

Notes: Conversion refers to purchases conditional on clicking. Percentages in the bottom panel are calculated among users who clicked at least one ad. The data confirm ordered search: users rarely click the second ad without first clicking the prominent ad.

D.2 Distributional Comparisons

Figure D.1 compares key variables for the same ad when displayed in a prominent versus non-prominent position. The plots show systematic differences in CPC, click-through rates, conversion rates, and cost-per-order (CPO), consistent with the prominence effect.

D.3 Temporal Patterns

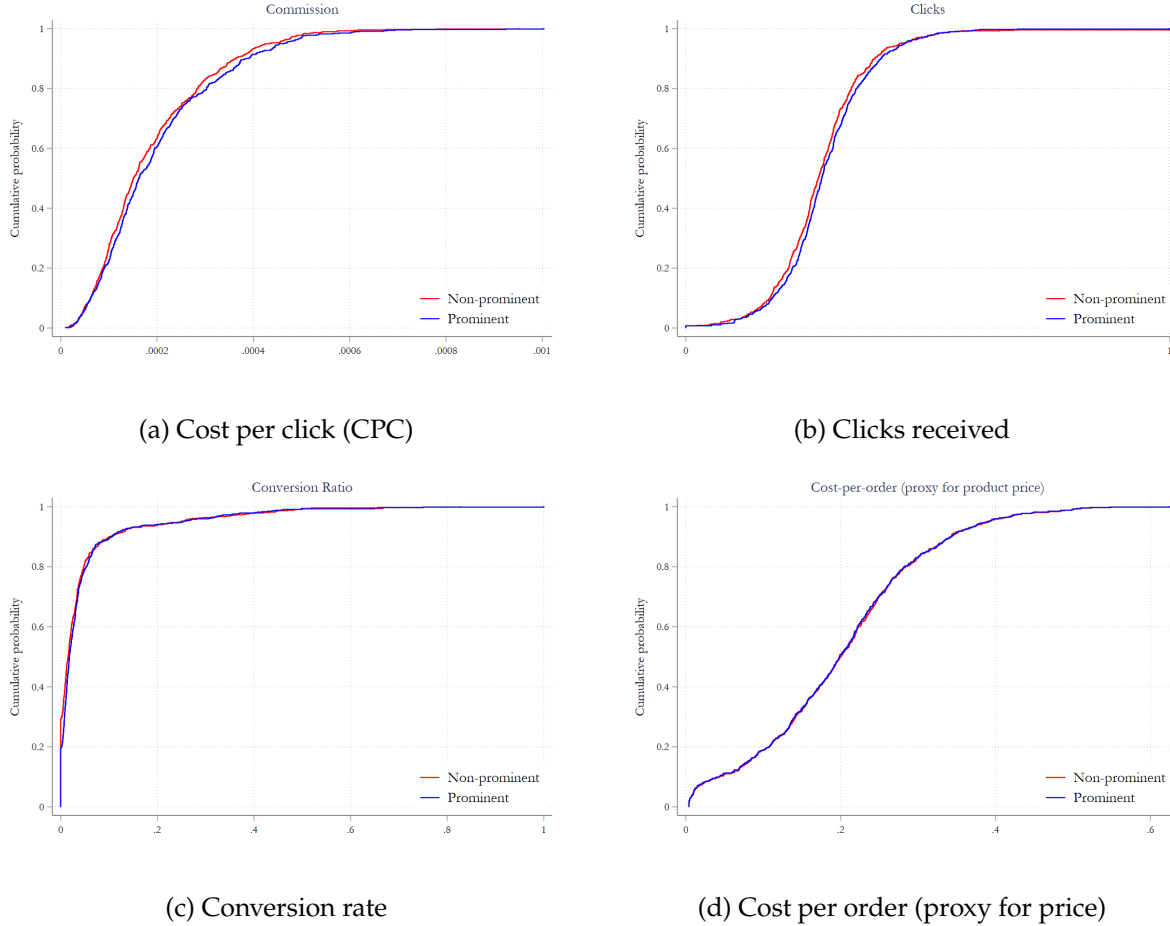
Figures D.2 and D.3 show daily and hourly patterns of search and purchase activity. Activity peaks during weekends and around 8 PM, consistent with prime-time engagement. These patterns motivate the inclusion of time controls in the empirical strategy.

D.4 Figures and Tables

Heterogeneity. As additional evidence, I hypothesise that prominence is more valuable when the demography is more likely to engage (in the form of higher clicks or purchases). For example, users visiting a website on “tips to purchase a wi-fi modem” are more valuable for a wi-fi service provider than users visiting other websites, as they are more likely to click and purchase from an ad on wi-fi. Conditional on observing higher total engagement over the two positions, Table D.2 confirms that the prominent firms pay a higher commission compared to the case when they appear with ad-pairs that receive low user engagement.

Evidence on bidders’ strategic interactions and equilibrium refinement. De Paula and Tang (2012) show that, under certain conditions, the second-order moments of equilibrium outcomes (commissions, in my setting) can be informative about the strategic interactions in the actions

Figure D.1: Cumulative distribution of key variables by ad position



Notes: Each panel shows the cumulative distribution of the variable for the same ad displayed in prominent versus non-prominent positions, controlling for ad fixed effects. CPC and cost-per-order are normalized by the data provider.

chosen (bids, in my setting) and therefore, about the presence of multiple equilibria in outcomes (commissions). More precisely, the sign of the moments can be informative about whether bids are strategic complements or substitutes. The key assumption required is that consumers' match values are independent across ads within a market, a standard assumption in consumer search frameworks that is also used in my theoretical model (see section 2).

Relying on this assumption, Table D.3 shows that firms' bids are strategic complements (in other words, there is a positive correlation between the bids of the firm that ends up being prominent and the firm that ends up being non-prominent), which provides support to the equilibrium refinement used in Section 4: of finding firms' bidding equilibrium in undominated strategies. To address concerns that this result might be driven by some highly valuable auctions, I

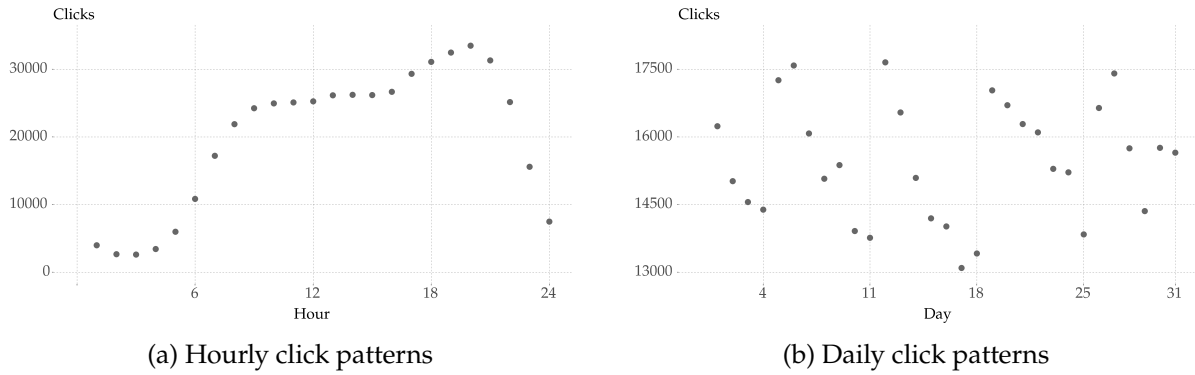


Figure D.2: Temporal variation in click behavior: Click activity peaks during evenings and weekends, consistent with prime-time engagement.

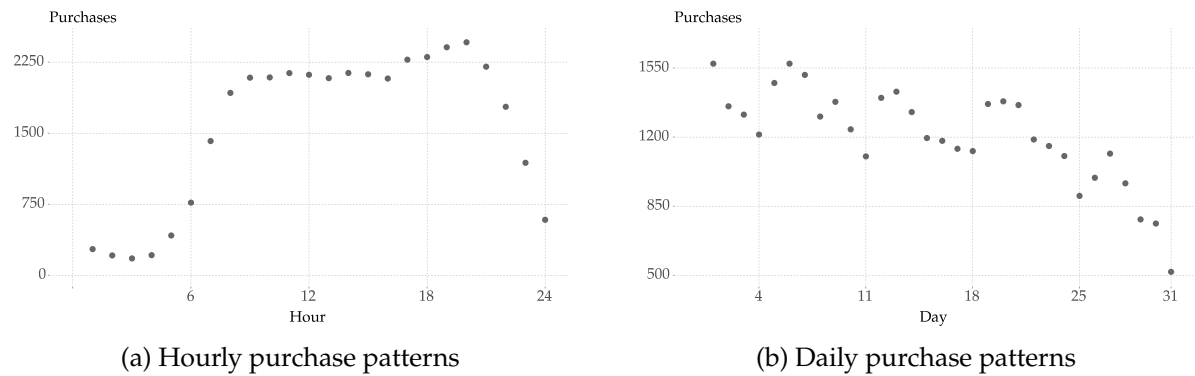


Figure D.3: Temporal variation in purchase behavior: Purchases follow similar cyclical patterns as clicks, with higher activity during evenings and weekends.

carry out the same exercise by focusing on a time period (the last week) with low user engagement (see Figure D.3). Table D.4 shows the baseline results hold even when user engagement is relatively lower, thus adding support to the interpretation that bidding strategies are, indeed, strategic complements.

Table D.2: Effect of prominence under heterogeneous user engagement

	(1)	(2)	(3)	(4)
Outcome: commission				
low user engagement (by clicks)	-0.000193*** (0.000)		-0.000143*** (0.000)	
low user engagement (by purchases)		-0.000329*** (0.000)		-0.000279*** (0.000)
prominence			5.11e-05*** (0.000)	9.49e-05* (0.000)
prominence $\times$ low user engagement (by clicks)			-9.03e-05*** (0.000)	
prominence $\times$ low user engagement (by purchases)				-9.30e-05* (0.000)
Observations	1,536,034	1,592,119	1,536,034	1,592,119
R-squared	0.052	0.051	0.052	0.051
mean(y)	0.000192	0.000192	0.000192	0.000192
sd(y)	0.000540	0.000540	0.000540	0.000540
ad FE	✓	✓	✓	✓
day of the week FE	✓	✓	✓	✓
hour of the day FE	✓	✓	✓	✓

Notes: The table reports estimates from regressions where the outcome is CPC. Columns (1)-(4) include ad fixed effects and time controls. Low user engagement is measured by clicks and purchases. Interaction terms capture whether prominence effects vary with engagement. Standard errors are in parentheses. *** denotes significance at 1%.

Table D.3: Evidence on equilibrium refinement: Strategic complementarity in bids

$\mathbb{E}[y_1 \cdot y_2]$	0.388*** (0.0003)
$\mathbb{E}[y_1] \cdot \mathbb{E}[y_2]$	0.286*** (0.0002)
$\mathbb{E}[y_1 \cdot y_2] - \mathbb{E}[y_1] \cdot \mathbb{E}[y_2]$	0.102*** (0.0004)

Notes: The table reports second-order moment tests for strategic interaction in bids. y_i denotes the equilibrium CPC for the firm in position i (1 = prominent, 2 = non-prominent). Controls include ad-pair, day-of-week, and hour-of-day fixed effects. The positive and significant difference between $\mathbb{E}[y_1 \cdot y_2]$ and $\mathbb{E}[y_1] \cdot \mathbb{E}[y_2]$ indicates that bids are strategic complements. Standard errors in parentheses; *** denotes significance at 1%.

Table D.4: Robustness: Strategic complementarity not driven by engagement levels

(a) Beginning of month (higher engagement)		(b) End of month (lower engagement)	
$\mathbb{E}[y_1 \cdot y_2]$	0.387*** (0.0003)	$\mathbb{E}[y_1 \cdot y_2]$	0.360*** (0.0006)
$\mathbb{E}[y_1] \cdot \mathbb{E}[y_2]$	0.293*** (0.0003)	$\mathbb{E}[y_1] \cdot \mathbb{E}[y_2]$	0.312*** (0.0007)
$sgn(\mathbb{E}[y_1 \cdot y_2] - \mathbb{E}[y_1] \cdot \mathbb{E}[y_2])$	0.094*** (0.0005)	$sgn(\mathbb{E}[y_1 \cdot y_2] - \mathbb{E}[y_1] \cdot \mathbb{E}[y_2])$	0.048*** (0.001)

Notes: The table reports second-order moment tests for strategic interaction in bids across periods of high and low user engagement. Engagement is proxied by purchase activity, which declines toward the end of the month (days 25-31), as shown in Figure D.3. Controls include ad-pair, day-of-week, and hour-of-day fixed effects. Positive and significant differences between $\mathbb{E}[y_1 \cdot y_2]$ and $\mathbb{E}[y_1] \cdot \mathbb{E}[y_2]$ in both panels confirm that bids remain strategic complements regardless of engagement level. Standard errors in parentheses; *** denotes significance at 1%.

For online publication: Appendix of *Sponsored Search and Product Market Concentration*

A Price Visibility

Equilibrium prices in a market where consumers are free to observe prices prior to visiting a firm's webpage (observable or posted) are lower than the equilibrium prices in the same market with unposted prices, *ceteris paribus*. Intuitively, a seller's demand is more elastic with respect to changes in posted prices than when price is only discovered after the consumer pays the search cost. Consider the unit mass of consumers plotted in Figure A.1. The co-ordinates of the square classify consumers by their valuation of the two products. For a small change in p_2 , we see that demand for firm 2 changes along both line segments LM and MN . Since firm 1 shares those line segments, these consumers are taken over by it. On the other hand, imagine a situation where prices are unposted. For a small change in p_2 , we see that demand for firm 2 changes along line segment LM only. In other words, demand for firm 2 is now less elastic compared to the case of posted prices. This reduction in elasticity is sufficient to overturn the order of prices.

In models of random search (Wolinsky, 1986, Anderson and Renault, 1999) and prominence (Armstrong, Vickers, and Zhou, 2009), firms might lower prices not to attract consumers, but to retain them once they visit. In my model, prices serve both roles: they help to both attract and retain consumers.^{A.1} Specifically, a lower price for firm 2 not only retains more visitors, but is also more likely to attract them; the latter effect makes its demand more elastic. This suggests that when consumers observe prices before searching, prices decrease with search costs. This can be observed from the price of firm 2. Since price of firm 1 is observed costlessly by the consumer at the beginning of search, this effect is absent in its case. Therefore, price dispersion increases with search cost in my model.

To further understand effect of posted prices, I disentangle the channels which influence firm 2's decision. When consumers visit firm 2, it learns that consumers, on average, have a lower valuation of product 1. This makes those consumers more likely to stay at firm 2, thus applying an upward force on p_2 . I refer to this as the *information channel*.

When prices are unposted, consumers visit firm 2 based on their rational belief about its price. However, it is free to surprise them as it doesn't change their incentives to visit firm 2. This also applies an upward force on p_2 . I refer to this as the *hold-up channel*. Note that, the case of unposted prices influences through both channels. However, making prices free to observe eliminates firm 2's ability to surprise and shuts down the *hold-up channel*. Therefore, I find that

^{A.1}Some previous studies consider prices as free to observe under different conditions. For instance, Dai (2017) uses it to analyse limited commitment and Arbatskaya (2007) uses it to analyse heterogeneous search cost.

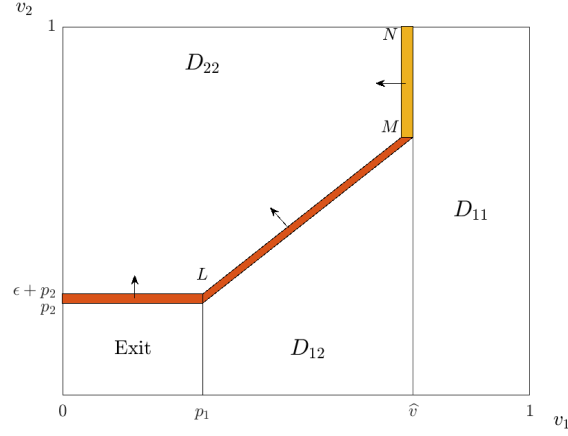


Figure A.1: Elasticity of demand: Intuition for prices

only having the information channel is not sufficient to generate increasing order of prices.

To illustrate this effect, I consider a market where $\lambda \in [0, 1]$ fraction of consumers can see both prices before starting their search while the rest discover prices by search. This exercise nests equilibrium prices from my benchmark model at $\lambda = 1$ and those from the [Armstrong, Vickers, and Zhou \(2009\)](#) (AVZ) model (which predicts $p_1 < p_2$) at $\lambda = 0$.

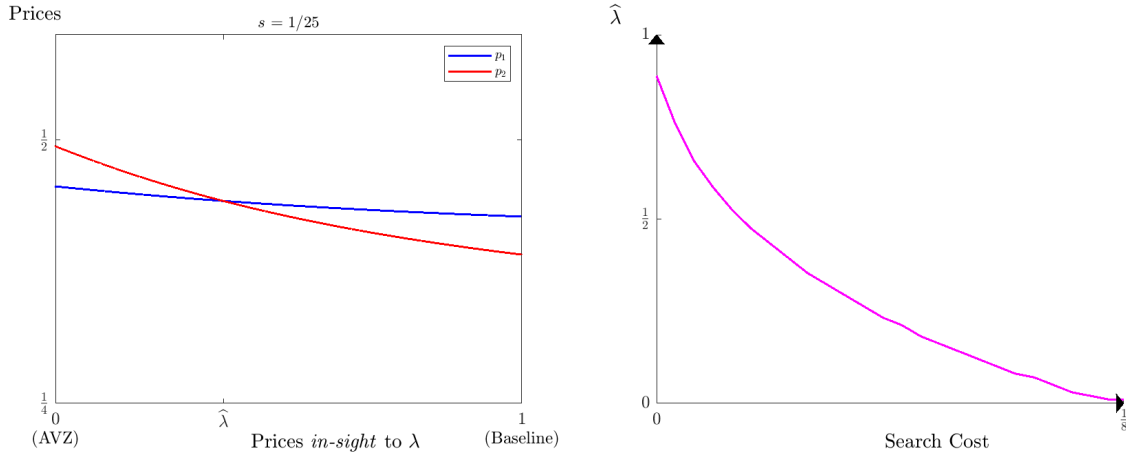


Figure A.2: Weakening of the Hold-up channel

Figure A.2 shows that as prices become free to observe to more consumers (increase in λ), the role of p_2 in attracting consumers gains significance. The *hold-up* force, which had caused higher prices for firm 2 in AVZ, is weaker now due to an increase in the price elasticity of firm 2's demand. This reverses the price order.

As search cost increases, it makes the firms incentives to attract consumers more important. Therefore, even when a smaller fraction of consumers see prices before search, we see a reversal in price rankings. Moreover, this exercise of parametrizing λ resembles the ‘clearing-house’ models of search (see, for instance, [Varian, 1980](#), [Perloff and Salop, 1985](#)) where a fraction of the population have access to prices. In those models, we usually see a mixed-strategy equilibrium. However, in my model, the underlying consumer heterogeneity allows for the existence of pure-strategy equilibrium.

B Revenues: ‘No Auction’

In this section, I derive the equilibrium revenues for both firms, in the absence of the auction.

Proposition B.1. *When match values are drawn from a Uniform distribution ($v \sim F = U[0, 1]$), the equilibrium revenue for firms diverge with search cost.*

$$\frac{dRev_1^*}{ds} > 0, \quad \frac{dRev_2^*}{ds} < 0$$

Proof. Using Envelope Theorem and substituting the expressions for the derivatives,

$$\begin{aligned} \frac{dRev_1^*}{ds} &= p_{1,NA}^* \left(\frac{dD_{11}^*}{ds} + \frac{dD_{12}^*}{ds} \right) \\ &= p_{1,NA}^* \left(-\frac{dp_{1,NA}^*}{ds} + (1 - p_{2,NA}^*) \frac{dp_{2,NA}^*}{ds} + 1 \right) \\ &= p_{1,NA}^* \left(-\frac{1 + (1 - p_{2,NA}^*) \frac{dp_{2,NA}^*}{ds}}{2} + (1 - p_{2,NA}^*) \frac{dp_{2,NA}^*}{ds} + 1 \right) \\ &= p_{1,NA}^* \left(\frac{1 + (1 - p_{2,NA}^*) \frac{dp_{2,NA}^*}{ds}}{2} \right) \\ &= p_{1,NA}^* \left(\frac{dp_{1,NA}^*}{ds} \right) \geq 0 \\ \frac{dRev_2^*}{ds} &= p_{2,NA}^* \left(\frac{dD_{22}^*}{ds} \right) \\ &= p_{2,NA}^* \left(\underbrace{-1 - (1 - p_{2,NA}^*) \frac{dp_{2,NA}^*}{ds}}_{\leq 1 \text{ from Proposition 1}} \right) \leq 0 \end{aligned}$$

■

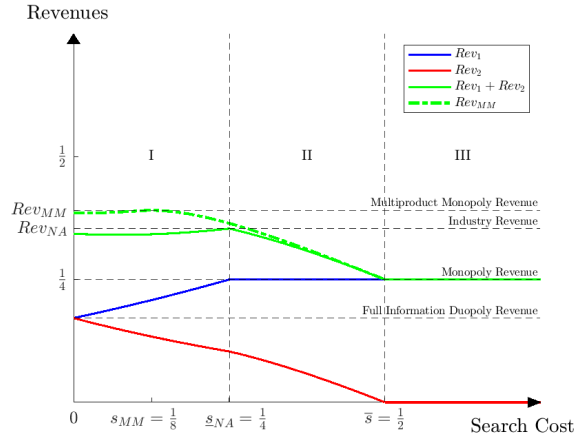


Figure B.1: Revenues (without auction)

Figure B.1 visualises the comparative statics of firm revenues, following proposition B.1. As search cost increases, there is a decrease in competition (region I). This increases the revenues as seen for firm 1 and reaches the monopoly level at $\underline{s}$. For $s > \underline{s}$, firm 1 maintains its optimal price at the monopoly level.

On the other hand, due to posted prices, firm 2 has lost its ability to extract the hold-up rent and the price decreases along with its (region I). In region II, firm 2's demand increases with s . This is because it now has a constant population of visitors to attract and retain and as it decreases its price, more consumers find the offer feasible and purchase. However, the revenues are well below the optimum due to constraint A. Therefore, revenue continues to decrease.

The industry revenue (sum of revenues) is maximum at the lowest search cost where both firms have segmented demand. This is because $\underline{s}$ would impose the loosest constraint on firm 2 while there is no competition (both firms have their own set of visitors).

Figure B.1 also compares the equilibrium industry revenues from my baseline model with the case where a single firm is selling both products. I refer to this second case as the Multi-product Monopoly (MM).^{B.1} Drawing parallels between the intermediary's function and MM can be helpful while considering policy tools applicable to the online market. From the perspective of the intermediary, this benchmark may also be useful to assess alternate position-allocation mechanisms.

The MM uses search cost to (imperfectly) screen consumers based on their preferences. Therefore, MM finds it optimal to set some $s > 0$. Only the consumers with lower valuations

^{B.1}See, for instance, Petrikaitė (2018) and Gamp (2016). Both papers characterise the optimal search cost such that a multi-product monopoly firm chooses to maximise its revenues.

for product 1 are catered to by the second product. This lets the MM charge a higher price for product 1, $(p_1)_{MM} > (p_2)_{MM}$. The MM does not lose as many consumers - if it raises its prices - as when firms are separate because some consumers buy from the other firm now (which is also a subsidiary of MM). Hence, prices of both products are higher in the case of MM than in my baseline model.

Figure B.1 also shows that the optimal s for the MM (s_{MM}) is lower than $\underline{s}$. This is because the MM is able to reach region II quicker. Since MM internalises the fact that some consumers that leave firm 1 for high prices buy product 2 instead, MM prices the goods more aggressively and there is a segmentation of the market for lower s . Thus, the maximum revenue for MM is also larger than the industry revenue when firms are separate.

C Constant Commission

In this section, I introduce to the firms' problem an ad cost or commission. I consider the case where firms face a constant commission for any search cost.

Firms' problem. Firms maximise their profits by choosing product prices ($\mathbf{p}$) and r_1 and r_2 are per-click commission to be paid by the respective firms. The analysis in Section 3 of the main text is a special case of this framework where $r_1 = r_2 = 0$. In Section 4 of the main text, r_1 and r_2 are endogenous objects derived from the auction. Firms maximise profit.

$$\max_{p_{1,C}} \pi_1 = p_{1,C}(D_{11} + D_{12}) - r_1(b_i, b_j, \widehat{b}) \quad (\text{C.1})$$

$$\max_{p_{2,C}} \pi_2 = p_{2,C}D_{22} - r_2(b_i, b_j, \widehat{b}) \cdot (1 - D_{11}) \quad (\text{C.2})$$

Subscript C denotes 'constant commission'. This framework with two firms parsimoniously allows me to highlight the differences in the cost structure of firms, due to positioning. From the above equation, we see the different roles per-click commissions play. Since all consumers visit firm 1, the ad commission plays the role of a fixed cost for firm 1, while it is analogous to a marginal cost per visitor for firm 2. This asymmetry, again a consequence of asymmetry in ad position, plays a crucial role in determining the equilibrium outcome, as shown in Proposition C.1 and later in the full equilibrium with auction in Theorem 2 of the main text.

Proposition C.1 (Constant commission). *Under an exogenous per-click commission fees of r_1 and*

r_2 is imposed, there is a unique asymmetric equilibrium of prices in pure strategies and are given by

$$p_{1,C}^* = \frac{1 - F(\widehat{v}^*) + \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p_C^*) \cdot f(v_1) \cdot dv_1}{f(\widehat{v}^*) - \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p_C^*) \cdot f'(v_1) \cdot dv_1}$$

$$p_{2,C}^* = \frac{A \cdot [1 - F(p_{2,C}^*)]F(p_{1,C}^*) + F(\widehat{v}^*) - F(p_1^*) - \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p_C^*) \cdot f(v_1) \cdot dv_1 + r_2 f(\widehat{v}^*)}{A \cdot f(p_{2,C}^*)F(p_{1,C}^*) + f(\widehat{v}^*) - F(p_2^*)f(p_1^*) - \int_{p_1^*}^{\widehat{v}^*} F(v_1 - \Delta p_C^*) \cdot f'(v_1) \cdot dv_1}$$

When match values are drawn from a Uniform distribution ($v \sim F = U[0, 1]$),

- the search cost at which the “Attraction” condition starts binding is lower than the ‘No auction’ case and is given by $\underline{s}_C = \frac{1-2r_2}{4} < \underline{s}_{NA}$
- the price of prominent firm increases with search cost $\frac{dp_{1,C}^*}{ds} > 0$, and the price of non-prominent firm decreases with search cost $\frac{dp_{2,C}^*}{ds} < 0$
- and there exists $\widehat{s}$ such that the price difference is positive for lower search costs, that is $p_{1,C}^* - p_{2,C}^* \begin{cases} < 0 & \text{if } s < \widehat{s} \\ > 0 & \text{if } s > \widehat{s} \end{cases}$

Proof. Prices are obtained by solving the first-order conditions for equation C.1. For $(p_1^*)_C, (p_2^*)_C$ to be global maxima:

$$p_1 \cdot \left(f'(\widehat{v}) - \int_{p_1}^{\widehat{v}} f(v_1 - \Delta p) \cdot f''(v_1) \cdot dv_1 \right) \leq 2 \left(f(\widehat{v}) - \int_{p_1}^{\widehat{v}} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right)$$

$$p_2 \cdot \left(A \cdot f'(p_{2,NA}^*)F(p_{1,NA}^*) - f'(\widehat{v}) - f(p_2)f(p_1) + F(\widehat{v} - \Delta p) \cdot f'(\widehat{v}) - \int_{p_1}^{\widehat{v}} f(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right)$$

$$\leq 2 \left(A \cdot f(p_{2,NA}^*)F(p_{1,NA}^*)f(\widehat{v}) - F(p_2)f(p_1) - \int_{p_1}^{\widehat{v}} F(v_1 - \Delta p) \cdot f'(v_1) \cdot dv_1 \right) - r_2 f'(\widehat{v})$$

At $s = 0$,

$$(p_1^*)_C = \sqrt{2} - 1 = p^{Ran}$$

$$(p_2^*)_C = \sqrt{2(1 + r_2)} - 1 > (p_1^*)_C$$

■

Proposition C.1 shows that for sufficiently low commission rates, there exist search costs for which $p_2 > p_1$ (see, for example, figure C.1). The marginal cost for visits for firm 2 increases its price which, in turn, gives room for firm 1 to also raise its price without the fear of losing

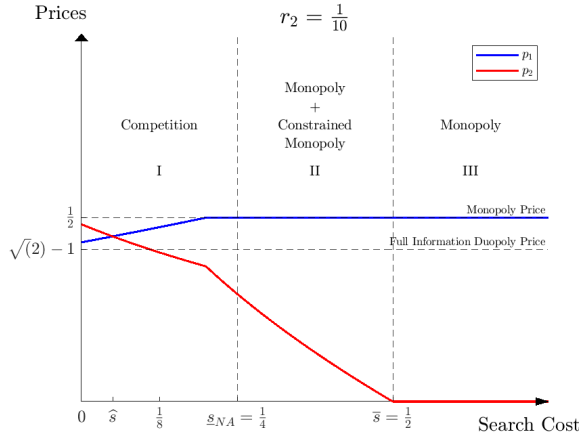


Figure C.1: Prices (with constant commission)

consumers. However, the effect on firm 1 is of second-order. Note that firm 1 faces a fixed cost of commission. Hence, there is an asymmetric rise in p_2 . This result reconciles with markets observed with increasing order of prices. However, the order is determined in my model due to the asymmetry in the cost structure. Thus, a non-prominent firm can charge a higher price, even when the *hold-up channel* has been eliminated.

The equilibrium prices diverge with search cost, following the properties highlighted in Section 3 of the main text. Note that $\underline{s}_C < \underline{s}_{NA}$ as prices are higher compared to the ‘No Auction’ case. The rise of p_2 due to marginal cost-like commission weakens competition as it damages firm 2’s ability to attract and retain consumers. This gives firm 1 more power in the market for all s . Hence, p_1 shifts closer to the monopoly price and leads to a segmented market for a smaller s .

D Costly first search

For $s > \bar{s}$ (region III), the market breaks down as nobody is interested in searching. The action happens in region II.

Recall that the reservation value for a consumer considering between staying at firm 1 and visiting firm 2 is given by

$$\hat{v} = 1 + p_1 - p_2 - \sqrt{2s}$$

The market would be active only if the reservation value exceeds the outside option. For $\underline{s} < s < \bar{s}$, this constraint binds (region II) and can be written as $p_2 < 1 - \sqrt{2s}$. The only question that remains now is to determine if the firms would prefer to deviate from $p_1 = p_2 = 1 - \sqrt{2s}$ (the equilibrium prices at $s = \underline{s}$) and profits of $\pi_1 = (1 - \sqrt{2s})\sqrt{2s}$ and $\pi_2 = (1 - \sqrt{2s})^2\sqrt{2s}$.

They do not deviate to a higher price because they would then not have any visitors. Let firm 1 charge $p_1 = 1 - \sqrt{2s}$ and firm 2 deviate to $p_2 = 1 - \sqrt{2s} - \epsilon$. Then, $\hat{v} > p_1$ and we are in the competitive framework (à la region I). Firm 1 gets a measure 1 of visitors and a measure $\sqrt{2s}$ of buyers among them. As an approximation, let me ignore the ϵ visitors that consider both firms, since their contribution to firms' demand would only be second-order. So, the remaining visit firm 2 and it gets a measure $(1 - \sqrt{2s})(\sqrt{2s} + \epsilon)$ of buyers. Then, firm 2 gets a profit of $(1 - \sqrt{2s})(1 - \sqrt{2s} - \epsilon)(\sqrt{2s} + \epsilon)$.

$$\pi_2 + (1 - 2\sqrt{2s} + 2s)\epsilon - (\sqrt{2s} - 2s)\epsilon - \epsilon^2(\cdot) \approx \pi_2 + (1 - 4\sqrt{2s} + 4s)\epsilon > \pi_2$$

Hence, it will prefer to deviate. Now, imagine firm 1 deviates instead. its profit would be $(1 - \sqrt{2s} - \epsilon)(\sqrt{2s} + D_{12})$ where $\epsilon > D_{12} = \mathbb{P}[v_1 > v_2 - \epsilon] = (\frac{1}{2} - \epsilon)$.

$$\pi_1 - \sqrt{2s}\epsilon + (\frac{1}{2} - \epsilon)(1 - \sqrt{2s} - \epsilon) \approx \pi_1 + \frac{1 - \sqrt{2s} - \epsilon}{2} - \epsilon > \pi_1$$

Hence, it also prefers to deviate. Now, imagine $p_1 = p_2 = 0$ and firm 2 deviates to $p_2 = \epsilon$. This would generate a positive profit for firm 2. Therefore, although it requires a more formal analysis, it seems like there is no equilibrium in pure-strategies for $\underline{s} < s < \bar{s}$.

E Timing of the game

In the baseline model, firms place their bids and learn their position before setting product prices. This is motivated by the intuition that the one-shot auction in my model is an approximation of a long dynamic game which has revealed all the unknown information about the auction outcome. Hence, firms set prices as if they know their position and commission. See also, for instance, [Chen, Mislove, and Wilson \(2016\)](#) and [Gorodnichenko and Talavera \(2017\)](#) for recent evidence on the high frequency of price changes in online markets. To extend the fit of my model to a more general setting, I explore alternate timings below.

Auction reserve price → firms bid and set prices simultaneously: This case is relevant to situations where auctions are nearly as frequent as product price adjustments. In symmetric equilibrium, the firms now have to set one price for both positions they will take up with equal

probability. In such situations, firms' objective can be represented by

$$\begin{aligned}
\max_{p_i} Rev_i &= p_i \cdot \frac{D_1 + D_2}{2} = p_i \cdot \frac{1 - p_i p_j}{2} \\
\text{Symmetry} \implies p &= \frac{1}{\sqrt{3}} (= p^{MM}) \\
\implies Rev &= \frac{1}{3\sqrt{3}} \approx 0.192 \\
\implies \hat{b} &= \frac{1}{\sqrt{3}} \\
\implies Rev_m &\approx 0.385 > \max\{Rev_m^{baseline}(s)\} \\
\pi &= 0
\end{aligned}$$

Note that this outcome is not a function of search cost. The asymmetric equilibria are identical to the baseline model, since we look at subgame-perfect Nash equilibria and hence, beliefs are true in equilibrium. However, we do not have the symmetric equilibrium we derived for the baseline model.

The crucial difference here is that firms do not know their position when they are setting a price. In the symmetric equilibrium of the baseline model, even though firms get each position with equal probability (due to tie-breaking), they know their position when they set prices. For example, consider that there are 100 instances of the same two firms competing for the top spot. In the baseline model, the position-allocation is revealed to the firms before the price setting. However, this is not true when firms set prices and bid simultaneously. In each of the 100 instances, the firms have to set prices without knowing their position, even though they end up getting each position with equal probability.

Auction reserve price → firms set prices → firms bid: This case is relevant to situations where auctions are rare (for e.g., quarterly) compared to product price adjustments. Solving backwards, we know that demand at position 1 is higher, even though they are more price elastic. Both firms prefer the first position and they both bid $b = Rev^{Mon}$. Therefore, they set prices equal to $\frac{1}{\sqrt{3}}$ since they do not know their position when they set prices. The logic from the previous case applies here as well.

Firms set prices → auction reserve price → firms bid: This case is relevant to situations where ad display designs changes at a higher frequency than product prices. In other words, pricing strategies are much more stickier than advertisement strategies.

The outcome here is same as in the previous case. Exchanging the timing of setting auction reserve price does not change outcomes for subgame-perfect Nash equilibria.

F Position auction

Equilibrium refinement: The tradition of restricting the set to undominated strategies is much older and attributed to analyses of co-operative game theory where the ‘core’ of a game is given by the undominated outcomes (Roth and Sotomayor, 1990). If as a planner or auctioneer, one would like to advice the firms to avoid some ‘bad strategies’, one may prescribe playing the undominated strategies. Note that this may not necessarily lead to a unique equilibrium.

Equilibrium in undominated strategies coincides with a unique locally-envy free refinement in my model. Edelman, Ostrovsky, and Schwarz (2007) develops this refinement informed by the markets’ dynamic structure. Although there is still multiplicity due to the fact that firm 1’s range of bids does not have an upper bound, this does not play a role in the equilibrium outcome.

Further, Varian (2007) also independently addresses the multiplicity and illustrates two possible refinements. The paper describes the intuition behind an aggressive approach - “what is the highest bid I can set so that if I happen to exceed the bid of the agent above me and I move up by one slot, I am sure to make at least as much profit as I make now?” - and a defensive approach - “If I set my bid too high, I will squeeze the profit of the player ahead of me so much that it might prefer to move down to my position”. The paper also derives a lower-bound of outcomes from a set of sub-game perfect Nash equilibria which coincides with the ‘locally envy-free’ refinement mentioned above.

Microfoundation: Standard models of position auction oversee a certain form of interdependency in object (position in this case) values across bidders (firms in this case). The value of a click for a firm is assumed to be constant across positions (see, for instance, Varian, 2009). Although it is common to assume that the click-through rate as a product of firm-specific factor times a position-specific factor, consumer behaviour remains simplified. For instance, Athey and Ellison (2011) assumes the number of clicks to be exogenous. These simplifications may not be innocuous but definitely helped gain insights on various features of the position auctions.

In my model, I derive consumer behaviour as a function of the price, which in turn, is a function of both value of a buyer and number of clicks and the commission (or the cost per click). Previous literature assumed the shape of these functions. Using consumer search as a model of consumer behaviour, we now know how these functions would look like. More importantly, now we know not just the functions but that there is a feedback, and what we want and can find is the fixed point.

Reserve Price: Edelman and Schwarz (2010) assesses the welfare effects of the auction reserve price. They separate the effects into direct (causing the lowest value bidder to face a higher payment - in my model it is firm 2) and indirect (inducing other bidders to increase their bids, thereby increasing others’ payments - in my model it is firm 1). They find that most of the

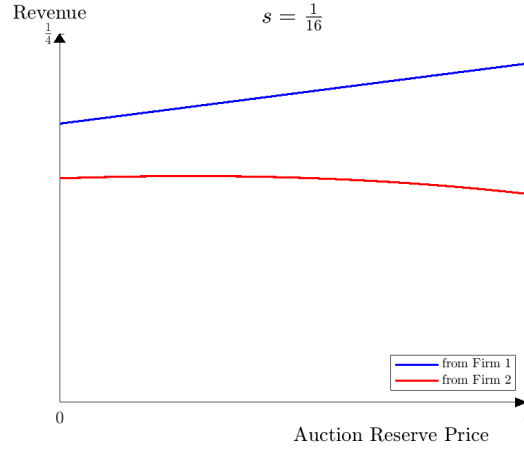


Figure F.1: Indirect effect of reserve price is stronger than direct effect

incremental revenue from setting a reserve price optimally comes not from a direct effect, but rather from the indirect effects on high bidders. I find a similar result that the larger share of intermediary revenue comes from firm 1 for any increment in $\hat{b}$ (see, for instance, figure F.1). Although top bidders' large valuations place them 'furthest' from the reserve price, they contribute more as commission. This is due to two forces. First, firm 2 is now forced to charge a higher price to cover its commission cost which allows firm 1 to also increase (relatively smaller than firm 2's increase) its price without loss. Second, this makes firm 2 less attractive. Thus, overall revenues of firm 1 increases substantially, to the benefit of the intermediary. Further, characterisation of the auction reserve price taking into account endogenous consumer behaviour has shown the existence of a strategy-proof and optimal allocation mechanism.

Similarly, on the other side, the auction reserve price also has an effect on the overall conversion rate in the market and how many consumers choose outside option.

One-shot auction: Note that the assumption of one-shot simultaneous bidding is a deviation from the possibility of continuous asynchronous bidding allowed in auctions run by some popular search engines. As in previous theoretical work on position auctions, I make this assumption for simplicity. [Edelman, Ostrovsky, and Schwarz \(2007\)](#) shows that the "lowest revenue envy-free" equilibrium generates the upper bound of revenues from dynamic auction models. However, having relaxed the assumption of homogenous position effects in my model, this may no longer clearly be the case and requires further study. This concern is also shared in [Goldman and Rao \(2016\)](#). Unfortunately, an analysis of this dynamic auctions is beyond the scope of this paper.

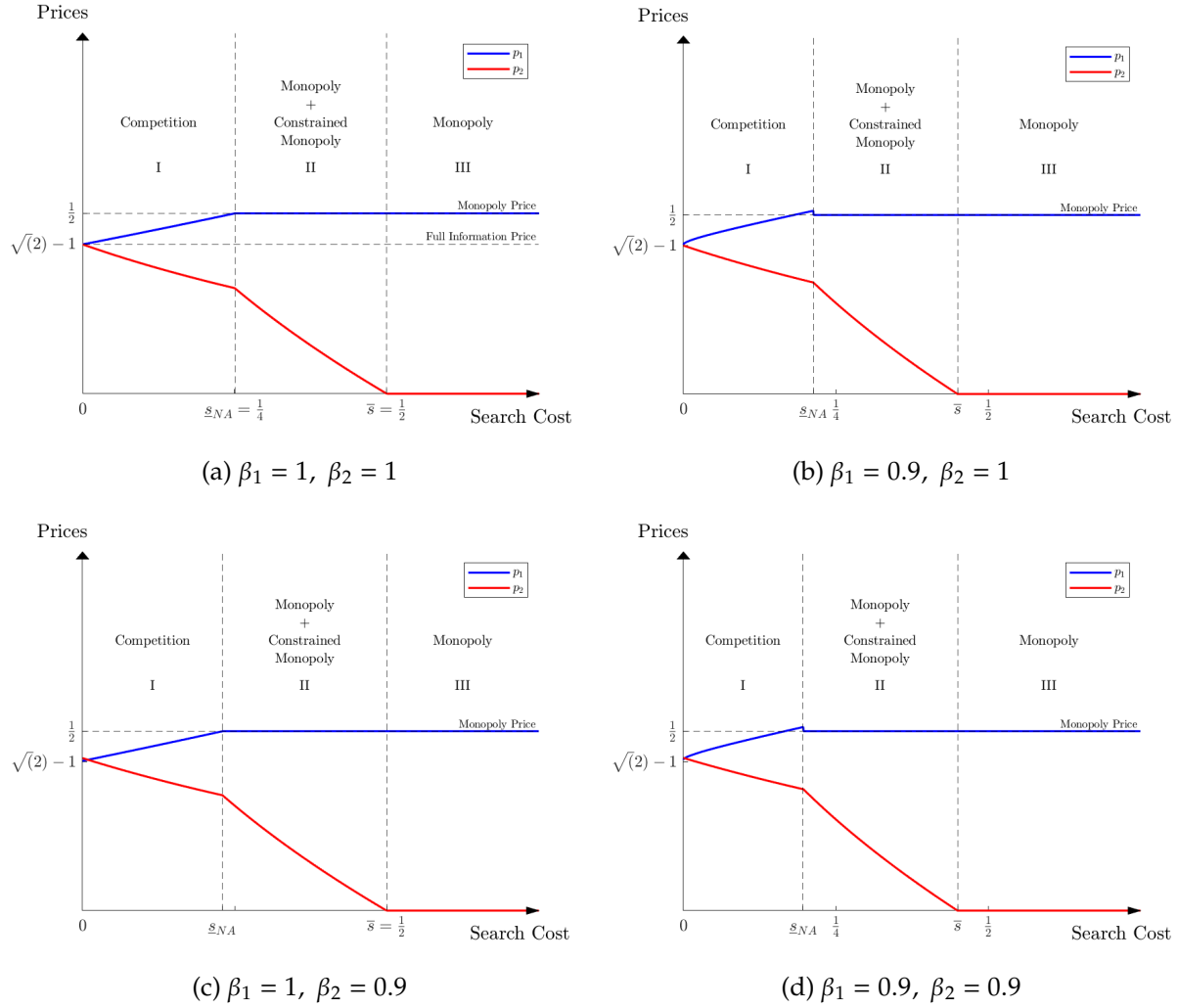


Figure G.1: Relevance: Without auction

G Asymmetric Firms: Branding/Relevance

Imagine a consumer searching for a particular brand as the keyword. Competitors of this ‘focal brand’ may also bid for ad positions. Consumers are more likely to visit these firms if their products are more relevant to their initial search.^{G.1} However, it is *ex-ante* unclear how many of these consumers the firms will be able to convert and how their equilibrium prices will be. One can also interpret this as the intermediary targeting its display ads to a demography that is

^{G.1}See, for instance, [Eliaz and Spiegel \(2011\)](#) and [Simonov and Hill \(2021\)](#).

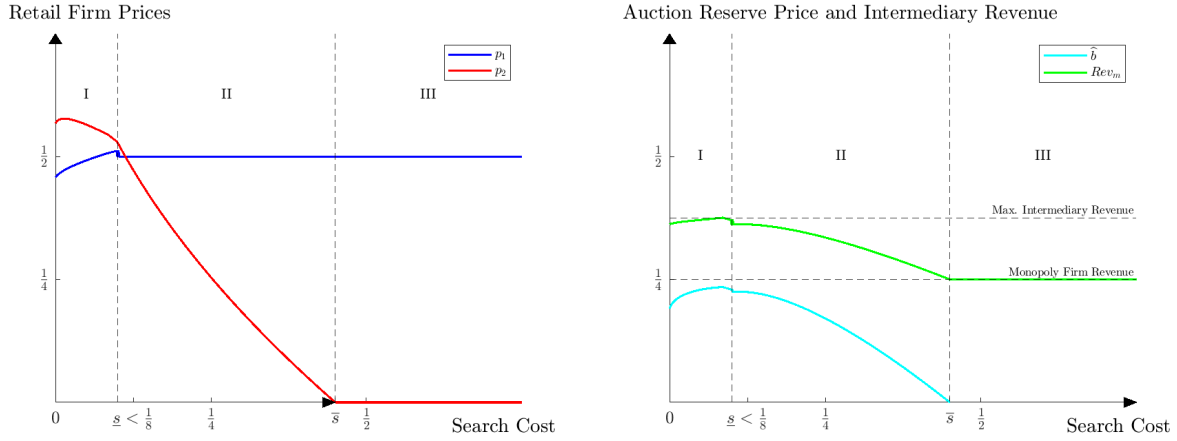


Figure G.2: Relevance ($\beta_1 = 1$, $\beta_2 = 0.9$): With auction

interested in a particular brand. I explore this extension of asymmetric firms in this section.^{G.2}

Let firms have relevance $\beta_1 > \beta_2$ (abuse of notation anticipates the equilibrium outcome). Let $\tilde{D}$ denote the demand for firms when there is heterogeneous relevance. The value of product $k = \{i, j\}$ (from firm k) for each consumer is drawn from a mixture

$$\begin{cases} = 0 & \text{with prob. } 1 - \beta_k \\ \sim U[0, 1] & \text{with prob. } \beta_k \end{cases}$$

Deriving the reservation value for the search rule, we get

$$\tilde{v}_\beta = \beta_2 + \tilde{\Delta p} - \sqrt{2\beta_2 s}$$

where $\tilde{\Delta p} = p_1 - \beta_2 p_2$. Note that the effect of p_2 on $\tilde{v}_\beta$ is dampened but β_2 is now an additional characteristic about firm 2 that consumers consider before deciding whether to search. Figure G.2 plots the equilibrium results for $F = U[0, 1]$.

^{G.2}Another way to think about this extension is that some fraction of consumers who search for a particular brand find a competitor brand's ad to be less relevant. This kind of heterogeneity in ordering by relevance might also happen when firms misunderstand the keyword yet, mistakenly, participate. On the other hand, a platform may even do this deliberately for two reasons. One, when they have seen many consumers who enter a certain keyword but end up buying a slightly different product, or second, when they want to increase the visibility of a certain firm. I do not take a stand on this and model the reduced form.

H Alternative Auction Format

Digital platforms are constantly innovating in their technology to sell ads, especially in the type of auction conducted.^{H.1} In the following exercise, I compare the baseline format of Generalised Second Price auction with the Generalised First Price auction (implemented widely by Yahoo!).^{H.2} I analyse the implications on welfare when the intermediary conducts an auction and both firms *pay their own bid* as the per-click ad commission.

Proposition H.1. *Symmetric pricing equilibrium does not exist (due to the tie-breaking rule). Asymmetric pricing equilibrium in pure strategies resembles that of the baseline model (Proposition 2 of the main text).*

Corollary H.1. *Following Proposition H.1, Proposition C.1 of the main appendix text on the equilibrium outcomes in a cost-per-purchase model also holds in a first-price auction setting.*

Firstly, there is non-existence of a symmetric equilibrium due to cyclic deviations: each firm would like to reduce its bid in order to pay less commission, but for a low enough competing bid, it would like to raise its bid to obtain the prominent position. Secondly, there is no difference between using first or second price auction from the firms' point of view, since under the symmetric bidding equilibrium, the highest bid and the second highest bid coincide. Since I resolve this symmetry using the tie-breaking rule to assign positions, the sub-game that follows is also identical.

^{H.1}See, for instance, CMA (2020a) and Ferrer, Ilango, and Richter (2022) for an overview of recent technological developments and Decarolis, Goldmanis, and Penta (2018) for a discussion of auction formats.

^{H.2}Note that for the case of two firms, a Vickrey-Clarke-Groves auction (implemented widely by Facebook) coincides, by design, with the GSP auction (implemented widely by Google).